

Optimal Human AI Coordination in Decision Workflows: Collaboration Paradox and Automation Cliffs

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As AI systems are integrated into organizational workflows, decision makers must determine how AI should be used for a given task: whether decisions are made by AI alone, by humans alone, or with AI assistance. We study this problem using a game-theoretic framework in which a coordinator commits to a randomized coordination policy to maximize system performance subject to limited human capacity, while human experts choose their level of involvement to minimize effort subject to a minimum performance requirement. This interaction gives rise to a generalized Nash equilibrium. In the setting with static AI performance, we characterize the equilibrium in closed form and show how human capacity and performance standards partition the design space into a set of interpretable regimes. Managerial implications are revealed from our analysis: (1) collaboration paradox: AI-assisted decision making need not be optimal even when it outperforms both human-only and AI-only operation on a per-task basis; and (2) automation cliffs: optimal coordination exhibits phase-change behavior as AI performance crosses key thresholds.

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1 Introduction

AI systems are rapidly expanding from narrow prediction tools into general purpose components of organizational workflows. These systems can now produce high quality recommendations, and in some settings executable actions, for diverse tasks such as triage [Yin et al., 2025], screening [Dai and Singh, 2024], routing [Liu et al., 2026], and manufacturing [Senoner et al., 2022]. As a result, organizations are increasingly embedding AI into operational processes. At the same time, the performance gains from AI assistance vary sharply across workflows. Prior work documents large improvements for tasks such as creative idea generation, market segmentation, drafting marketing copy, and persuasive memos, while showing that AI can harm performance on tasks requiring careful integration of quantitative data and subtle qualitative evidence [Dell’Acqua et al., 2023]. This heterogeneity makes a single, uniform deployment mode untenable. Instead, across a large portfolio of organizational workflows, decision makers must determine where an AI system should act autonomously, where it should provide advice, and where humans should retain full control.

In this paper, we study this problem and characterize the optimal mode of AI integration for a given workflow. We focus on three common integration modes that arise in practice. Under *AI-only*, the system produces the decision without human involvement. Under *human-only*, the human expert produces the decision end to end without AI involvement. Under *AI-assisted* decision making, the AI provides a recommendation while the human retains final decision rights.

We characterize the optimal integration mode for a workflow along two dimensions. The first dimension is the workflow’s performance requirement. Each workflow must satisfy a minimum standard on outcomes, and the optimal integration mode depends on how this standard compares with performance under AI-only, human-only, and AI-assisted operation. The second dimension is human resources. Human involvement is capacity constrained and costly. Moreover, human experts are not passive components. When workload increases, experts devote less attention to reviewing AI outputs and may fail to detect errors. Evidence from the human factors literature shows that operators often insufficiently monitor or verify automated outputs under workload and time pressure [Parasuraman and Manzey, 2010, Parasuraman and Riley, 1997]. These considerations imply that AI and human decision making are not symmetric, and a realistic model must account for endogenous human effort.

Motivated by these observations, we propose a game theoretic model of human–AI coordination for a single workflow receiving a stream of homogeneous tasks. Tasks arrive over time as independent draws from an underlying population. Prior to operation, a coordinator commits to a randomized coordination policy that specifies the probabilities with which an arriving task is routed to AI-only, human-only, or AI-assisted human operation. The coordinator does not observe outcomes when choosing the policy; rather, it designs a routing rule for the workflow that will be applied repeatedly to many task instances. The coordinator chooses this policy to maximize expected system performance subject to a hard constraint on total human involvement, capturing limited human capacity. The human expert takes the routing policy as given and, whenever it is assigned a task, chooses an effort level that affects decision quality and the effectiveness of review. The expert minimizes effort cost subject to satisfying the workflow’s performance requirement. Because the coordinator’s routing policy determines the expert’s workload and the expert’s effort determines whether the performance requirement can be met under that workload, feasibility for each side depends on the other’s decision. The resulting solution is therefore a generalized Nash equilibrium that jointly determines the optimal coordination decision and the amount of human effort exerted.

We first analyze a static setting in which AI performance is fixed. The model admits a closed-form characterization of equilibrium coordination. The two key dimensions, workflow’s performance

requirement and the available human capacity budget, partition the design space into six regimes. The analysis yields two central implications:

- (1) *Collaboration Paradox*: Human–AI collaboration need not be optimal even when collaboration strictly dominates both human-only and AI-only modes in per-task performance and cost. The reason is strategic human engagement. When baseline AI performance approaches or exceeds the required standard, routing tasks to AI-assisted review reduces the marginal value of human effort. Anticipating that minimal intervention suffices to satisfy the performance constraint, human experts optimally exert low effort in reviewing AI outputs. This reduction in attention can offset the theoretical gains from collaboration. Empirical evidence documents similar patterns of automation complacency, where operators under-monitor highly reliable systems and fail to detect errors [Dell’Acqua, 2022, Parasuraman and Riley, 1997]. In equilibrium, therefore, collaboration improves system-level performance only when the required standard exceeds what AI alone can achieve *and* when human capacity is sufficiently limited that allocating a meaningful share of tasks to human-only processing is infeasible.
- (2) *Automation Cliffs*: The optimal coordination policy exhibits sharp phase transitions as AI performance crosses critical thresholds. When AI accuracy remains below human performance, the full spectrum of integration modes can arise in equilibrium, including human only operation, parallel use of humans and AI without interaction, partial collaboration, and full AI assisted review. In this region, small improvements in AI capability typically induce only marginal adjustments in routing probabilities. In contrast, once AI performance surpasses the human benchmark, the equilibrium structure changes discontinuously. Regions in which human-only operation or non-collaborative modes were previously optimal collapse, and tasks are reassigned to AI-only operation. Managerially, this implies that the optimal level of human involvement is not a smooth function of AI quality. Instead, organizations should expect tipping point dynamics in which incremental technical improvements trigger large scale substitution of automation for human effort. Consequently, the organizational impact of AI may appear muted until performance crosses a critical boundary, after which adoption accelerates rapidly and human roles are reconfigured at scale.

The static analysis treats AI quality as exogenous. In practice, coordination determines the supervision the system receives and therefore its subsequent performance. To incorporate this feedback, we extend the model in Appendix B to a multi-stage environment in which the state variable is AI quality and routing decisions determine its law of motion. Each period, the coordinator selects a routing policy subject to the performance requirement and the human capacity constraint. The induced allocation governs both current outcomes and the quantity and composition of labeled data generated for learning. The dynamic problem therefore trades off contemporaneous feasibility against improvements in future AI capability. The core insights of the collaboration paradox and the automation cliffs persist in this dynamic setting. At the same time, learning reshapes early-stage allocations. Human involvement becomes more concentrated. Fewer tasks are routed to humans, but those that are reviewed receive higher per-task effort, increasing the effective intensity of supervision. Oversight is directed toward tasks where AI performance is close to the human benchmark and marginal learning returns are highest. As AI improves, the equilibrium path converges toward AI-only operation.

To summarize, the paper makes three contributions:

- (1) We develop a tractable analytical framework for studying how organizations allocate decision authority between humans and AI when human attention is scarce and effort is endogenous. The model highlights that coordination decisions must account not only for the comparative

accuracy of humans and AI, but also for how human engagement responds to workload and incentives.

- (2) We provide a structural characterization of equilibrium coordination regimes. The analysis identifies when automation, human control, or selective collaboration arise and formalizes two central phenomena: a collaboration paradox, in which endogenous disengagement undermines assisted review, and automation cliffs, in which small improvements in AI quality induce discontinuous shifts in task allocation.
- (3) Our work contributes to the literature on human–AI interaction by shifting the unit of analysis from individual predictions to workflow tasks. As far as we are aware, this paper is among the first to study human–AI coordination at the workflow level taking into account endogenous human effort, and to connect these coordination decisions to the dynamics of AI learning over time.

1.1 Literature Review

Our paper relates to several streams on human–AI delegation, AI assistance, and feedback-driven learning. Across these literatures, a common modeling choice is to take one side of the interaction (human behavior, the institution’s policy, or the AI learning process) as exogenous. Our contribution is complementary: we study how a coordinator’s routing policy and a human expert’s endogenous effort jointly determine both contemporaneous performance and (in the multi-stage model) the evolution of AI quality.

A first line of research studies task allocation between humans and AI, often framed as delegation or deferral. This literature develops rules that assign each task either to an algorithm or to a human expert in order to optimize predictive accuracy or utility [Athey et al., 2020, Mozannar et al., 2023, Mozannar and Sontag, 2020, Wilder et al., 2021]. In these models, the system selects between human-only and AI-only decisions, and the human’s performance is typically treated as fixed. The focus is on learning optimal routing policies given exogenous performance characteristics. Our setting differs along two dimensions. We introduce AI-assisted review as a distinct mode in which the human observes and potentially overrides an AI recommendation, and we allow the human’s effort in this review to depend on the volume of assigned tasks. As a result, delegation decisions affect not only who makes the decision, but also how intensively humans engage with AI outputs.

A second stream studies AI assistance under a fixed interaction structure. In these models, the AI generates recommendations, feature highlights, or candidate sets, and the objective is to design the content of advice to improve human decisions [Bastani et al., 2026, Li and Zhu, 2024, Noti et al., 2025]. The interaction protocol is taken as given: tasks are assisted, and the allocation of decision authority does not vary across instances. Human response is either modeled behaviorally at the task level or treated as fixed. In our framework, the interaction structure itself is a choice variable. The coordinator selects routing probabilities across AI-only, human-only, and AI-assisted modes subject to a binding oversight-capacity constraint. AI-assistance is therefore not taken as given. Moreover, human effort in assisted review is chosen in response to workload and the required performance standard. As a result, the fraction of assisted tasks and the intensity of review are jointly determined in equilibrium. Coordination decisions alter incentives, and incentives feed back into system performance. This coupling of routing and endogenous effort distinguishes our setting from models that optimize recommendation design within a fixed protocol.

A complementary literature examines human supervision, belief updating, and feedback loops in high-stakes AI systems [Balakrishnan et al., 2026, De Véricourt and Gurkan, 2023, Ibrahim et al., 2021, Verma et al., 2025, Zhou et al., 2025]. These works highlight how human feedback affects reliability and learning. We formalize this connection in a dynamic environment in which routing

decisions determine the quantity and composition of supervision and thereby the law of motion of AI quality.

More broadly, interactive human–algorithm systems generate strategic and dynamic responses [Castro et al., 2025, Delarue et al., 2025, Fu et al., 2025]. Closely related, [McLaughlin and Spiess, 2024] adopt a principal–agent perspective in which human behavior adapts to algorithmic policies. Our framework shares this focus on strategic adaptation but differs in two respects. We allow randomized triage across three integration modes rather than a binary delegation choice, and we embed this allocation problem in a multi-stage learning environment in which current routing decisions affect future AI performance.

Taken together, these literatures motivate the need for models that integrate institutional delegation under capacity constraints, endogenous human engagement under performance standards, and dynamic learning effects. Our framework provides such an integrated and tractable workflow-level perspective.

2 Core Model: Static Coordination

2.1 Problem Setup

We study a static coordination problem in which learning is absent¹ and the algorithmic policy is fixed. Tasks of a single type arrive over time as independent draws from an underlying population. For each task, a coordinator commits to a randomized coordination policy $p = (p_A, p_H, p_R)$, where p_m denotes the probability of assigning a task to modality $m \in \{A, H, R\}$ and $p_A + p_H + p_R = 1$. The three modalities correspond to full automation by the algorithm, full human decision making without algorithmic input, and assisted decision making in which the AI issues a recommendation that the human may review. As illustrated by Fig. 1, we now describe each modality in detail:

- (1) *Full automation (A)*. When the coordinator delegates a task to the AI alone, the task is executed automatically and yields expected reward r_A . We do not include a per-task algorithmic cost; in many operational settings marginal inference costs are negligible relative to constrained human attention.
- (2) *Human-only decision making (H)*. When the coordinator routes a task to the human without algorithmic input, the expected reward is r_H and the human incurs effort cost c_H per task. The cost c_H captures time, cognitive load, and workflow disruption associated with producing an output without algorithmic assistance.
- (3) *Assisted decision making (R)*. When the coordinator routes a task to assisted decision making, the AI produces a recommendation that is shown to the human. Conditional on receiving a recommendation, the human chooses whether to engage by verifying, editing, or overriding the suggestion. We denote by $q_R \in [0, 1]$ the probability that the human engages when a recommendation is presented. If the human engages, the realized reward is r_R and the human incurs effort cost c_R . If the human does not engage, the recommendation is accepted and the realized reward is r_A . When assisted decision making is used with probability p_R , the expected reward contribution from that modality is $p_R(r_R q_R + r_A(1 - q_R))$ and the expected human effort consumed by that modality is $p_R q_R c_R$.

Each of these three modes is well studied in the human–AI collaboration literature. Relative to standard delegation models, we assume that when assisted decision making is used, the organization can decide whether an AI recommendation is provided, but the intensity of review is chosen by the human. We represent this discretion by an engagement probability q_R . We model the human as choosing the least costly level of engagement that satisfies an institutional performance requirement,

¹The extension that considers dynamic learning effects of multi-stage human-AI coordination is in Appendix B.

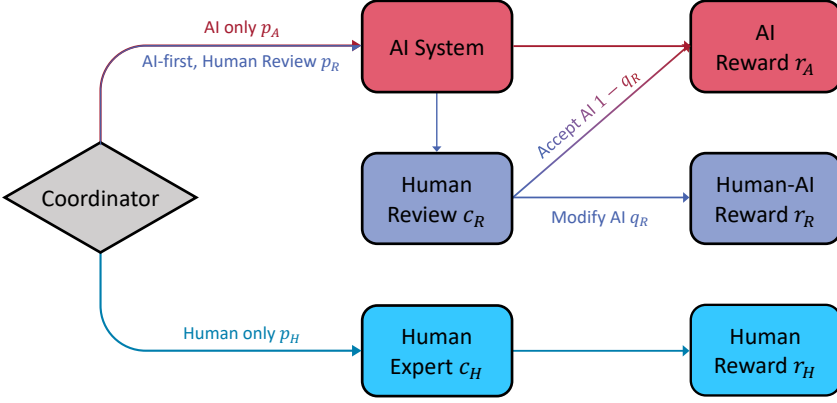


Fig. 1. Illustration of the human and coordinator policies.

reflecting that review effort is costly and typically not directly observable, while performance is evaluated at an aggregate level. Accordingly, organizations enforce outcome standards rather than prescribing or monitoring specific review actions [Holmström, 1979, Sherstyuk, 2000]. Specifically, the human behavior model can be written as:

$$\begin{aligned} \min_{q_R \in [0,1]} \quad & c_R p_R q_R \\ \text{s.t.} \quad & r_H p_H + r_A p_A + (r_R q_R + r_A(1 - q_R)) p_R \geq \delta. \end{aligned} \quad (1)$$

where the parameter δ denotes the institutionally required level of average performance.

Subject to the human behavior, the coordinator aims to choose a human-AI coordination strategy that maximizes the expected reward subject to a finite amount of human effort:

$$\begin{aligned} \max_{p \geq 0} \quad & r_H p_H + r_A p_A + (r_R q_R + r_A(1 - q_R)) p_R \\ \text{s.t.} \quad & c_H p_H + c_R p_R q_R \leq b, \\ & p_A + p_H + p_R = 1. \end{aligned} \quad (2)$$

The constraints in (1) and (2) determine which coordination policies are feasible and govern the equilibrium behavior that follows. Because the coordinator's feasible set depends on the human engagement probability through the effort budget, and the human's feasible set depends on the coordinator's mix of modalities through the performance requirement, the two optimization problems cannot be solved in isolation. We therefore formulate the interaction as a generalized Nash game [Arrow and Debreu, 1954, Debreu, 1952] in which each player optimizes subject to constraint sets that depend on the other player's choice. Proposition 2.1 records a simple sufficient condition for existence, which is proved in Appendix A.1.1.

PROPOSITION 2.1 (EXISTENCE OF EQUILIBRIUM FOR THE CORE (SINGLE-STAGE) MODEL). *Suppose that (i) for every $q_R \in [0, 1]$ feasible for (1), the coordinator's problem (2) has a nonempty feasible set; and (ii) for every p feasible for (2), the human problem (1) has a nonempty feasible set. Then the generalized Nash game induced by (1) and (2) admits at least one equilibrium.*

We now characterize equilibrium behavior. To keep the exposition compact and the equilibrium regions interpretable, we impose a weak complementarity condition that reflects common features of assisted decision making.

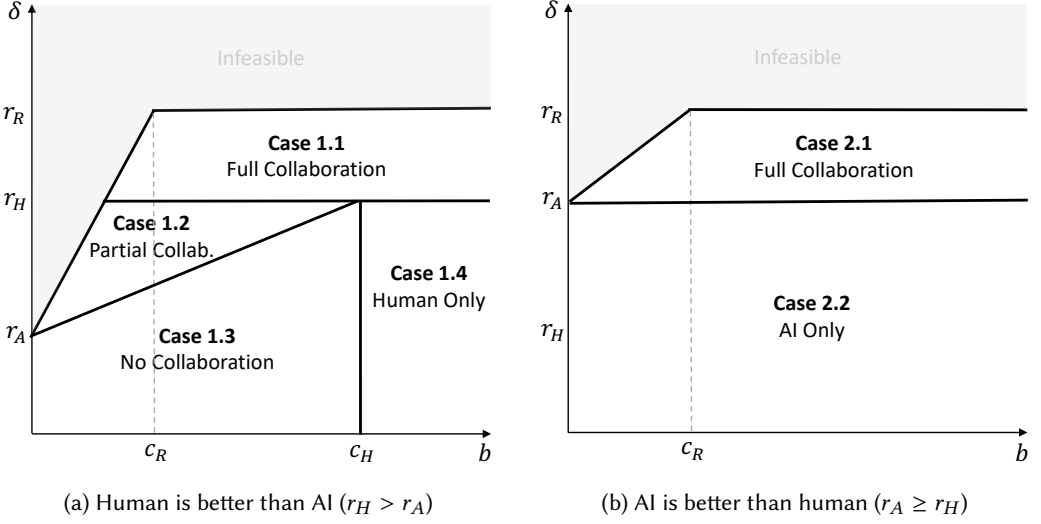


Fig. 2. Equilibrium regions in the core (single-stage) model.

ASSUMPTION 1 (COMPLEMENTARITY AND EDITING ECONOMY).

$$r_R \geq r_H, \quad r_R \geq r_A, \quad c_R \leq c_H.$$

Assumption 1 makes two assumptions. First, when the human reviews an AI recommendation, assisted decision making weakly improves outcomes relative to either agent acting alone. This is consistent with a growing body of evidence which shows the benefit for human-AI collaboration across different settings [Vaccaro et al., 2024]. Second, conditional on engagement, reviewing and editing an AI recommendation is no more costly than producing an output from scratch. This reflects experimental and field evidence that AI assistance often reduces labor input per task in writing and related production workflows [Noy and Zhang, 2023].

2.2 Analysis of Static Policies

Given the assumptions above, we can fully characterize the equilibrium for the core model:

THEOREM 2.2 (EQUILIBRIUM CHARACTERIZATION FOR THE CORE (SINGLE-STAGE) MODEL). *Suppose Assumption 1 and the feasibility conditions in Proposition 2.1 hold. Then the generalized Nash game induced by (1) and (2) admits an equilibrium. Moreover, the equilibrium policies (p^*, q^*) are piecewise defined over regions of the (b, δ) plane. When $r_H > r_A$, the regions and corresponding policies are summarized in Fig. 2a. When $r_A \geq r_H$, the regions and corresponding policies are summarized in Fig. 2b.*

The proof of Theorem 2.2 is contained in Appendices A.1.3, A.2, and A.3. Fig. 3 and 4 summarize the equilibrium structure of the core (single-stage) model. When the AI is at least as capable as the human ($r_A \geq r_H$), the feasible set is primarily governed by whether the requirement exceeds the AI baseline; see Fig. 4. For low and moderate values of δ , the coordinator assigns tasks almost entirely to the AI, with human oversight appearing only when the requirement approaches the reliability threshold r_R . As δ tightens further, human review arises only within a narrow wedge near the feasibility boundary, and beyond this boundary the problem becomes infeasible regardless of the available budget. In contrast, when the human outperforms the AI

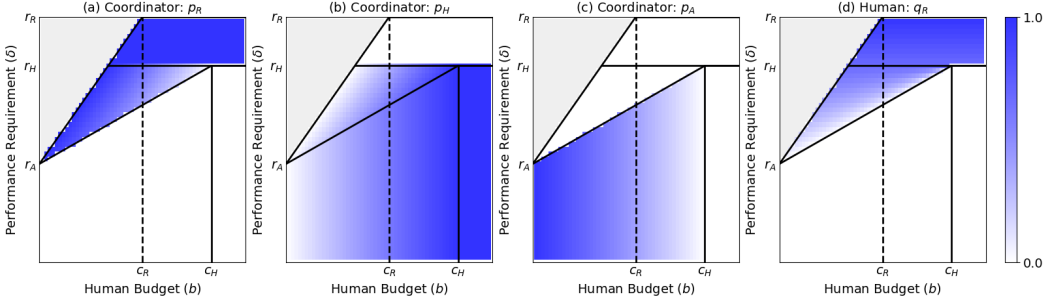


Fig. 3. Static policies in the core (single-stage) model when human is better than AI ($r_H > r_A$).

($r_H > r_A$), the equilibrium exhibits a richer structure, as shown in Fig. 3. With limited budget, stringent performance requirements can only be satisfied through direct human involvement, leading to regions in which the coordinator routes tasks to the AI/human or relies heavily on review. As the budget increases past the critical threshold C_R , selective collaboration emerges: the coordinator combines AI deployment with targeted human oversight, and the human’s best response concentrates effort on tasks where oversight is most valuable. Across both figures, the black boundaries delineate feasibility and regime transitions, illustrating that performance standards and oversight capacity jointly determine whether the equilibrium features full automation, full human control, or selective human–AI collaboration. We now formally describe each region.

2.2.1 Infeasible Region. A pair (b, δ) is infeasible if no combination of delegation and engagement can simultaneously satisfy the coordinator’s human-effort budget and the performance requirement. Because expected reward is bounded above by the best achievable assisted outcome, the most stringent requirement the system can ever satisfy is r_R . Thus, a necessary condition for feasibility is

$$\delta \leq r_R. \quad (3)$$

Moreover, even when $\delta \leq r_R$, meeting the requirement may demand a strictly positive level of engagement under assistance, which in turn consumes human budget. In particular, if the coordinator assigns all tasks to R (the best-performing modality under Assumption 1), the human must engage at least

$$q_{\min}(\delta) = \frac{\delta - r_A}{r_R - r_A} \quad (\text{interpreting } q_{\min} = 0 \text{ when } \delta \leq r_A)$$

to achieve average reward δ . This requires human cost $C_R q_{\min}(\delta)$ per task since, for $(p_A, p_H, p_R) = (0, 0, 1)$, we have that $r_R p_R q_{\min}(\delta) + r_A p_R (1 - q_{\min}(\delta)) + r_H p_H + r_A p_A = \delta$. Therefore, a necessary feasibility condition is

$$b \geq C_R q_{\min}(\delta) \quad (4)$$

When either Equation (3) or (4) fail, we cannot meet the required performance within the available human effort budget regardless of how the human responds. More details on how the infeasible region is derived appear in Appendix A.1.2. Operationally, it corresponds to settings where regulators or administrators demand a level of quality that would require more human review than is available. In such regions, the appropriate response is not a different equilibrium policy but rather a redesign of requirements (lower δ) or resources (increase b).

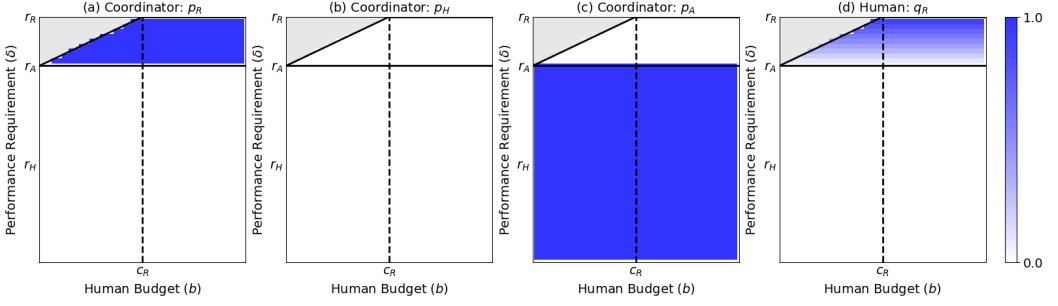


Fig. 4. Static policies in the core (single-stage) model when AI is better than human ($r_A \geq r_H$).

2.2.2 AI-Only. In the AI-only regime, all tasks are delegated to the AI and the human remains idle. The equilibrium policy is

$$p_A^* = 1, \quad p_H^* = 0, \quad p_R^* = 0, \quad q_R^* = 0.$$

This regime arises whenever the AI baseline (i) is better than human and (ii) already meets the institutional requirement,

$$r_A \geq r_H \quad \text{and} \quad \delta \leq r_A,$$

Because the AI already meets the institutional requirement, assigning tasks for review does not generate meaningful oversight in equilibrium. Instead, the human will choose to follow the recommendation with minimal scrutiny. This regime corresponds to Case 2.2 in Fig. 2b.

2.2.3 Human-Only. In the human-only regime, all tasks are handled by the human without any AI input:

$$p_H^* = 1, \quad p_A^* = 0, \quad p_R^* = 0, \quad q_R^* = 0.$$

This regime requires that (i) human is better than AI, (ii) human-only performance is sufficient to satisfy the standard, and (iii) that the budget can support full human processing:

$$r_H > r_A, \quad \delta \leq r_H, \quad \text{and} \quad b \geq c_H.$$

When these conditions hold, the coordinator can implement the performance requirement with H alone. Moreover, when $r_H > r_A$ (the left panel in Fig. 2), the coordinator prefers H to A on reward grounds; when $r_A \geq r_H$ (the right panel in Fig. 2), this regime is absent because H is weakly dominated by A in reward and strictly more costly.

Human-only corresponds to settings where institutions deliberately withhold AI recommendations even when they exist. The model predicts that such “AI silence” is sustainable as an equilibrium when (a) performance standards are achievable through human work alone and (b) the organization can afford the necessary human effort. This regime corresponds to Case 1.4 in Fig. 2a.

2.2.4 No-Collaboration. In the no-collaboration regime, the coordinator mixes between humans (H) and AI (A), but never uses assisted decision making (R). The equilibrium policy is

$$p_R^* = 0, \quad p_H^* = \frac{b}{c_H}, \quad p_A^* = 1 - \frac{b}{c_H}, \quad q_R^* = 0.$$

This regime arises when the human outperforms the AI, the performance requirement can be met without assistance, and the available human budget is insufficient to assign all tasks to the human:

$$r_H > r_A, \quad \delta \leq r_A + \frac{b}{c_H} \times (r_H - r_A), \quad \text{and} \quad b < c_H.$$

In this case, the coordinator can afford to assign only a fraction b/c_H of tasks to the human. Because the performance requirement satisfies $\delta \leq r_A + \frac{b}{c_H}(r_H - r_A)$, the system can meet the target by allocating sufficient mass directly to human-only decisions. Assisted decision making is therefore unnecessary from a feasibility standpoint. Moreover, using human effort for assisted review is suboptimal: given the limited budget, allocating effort to editing algorithmic recommendations delivers less performance improvement per unit of human cost than assigning those tasks entirely to the human. This regime corresponds to Case 1.3 in Fig. 2a.

2.2.5 Partial Collaboration. In the *partial collaboration* regime, the coordinator uses both assisted decision making (R) and the human alone (H). This region exists when the following three conditions hold:

$$r_H > r_A, \quad b \geq c_R q_{\min}^*(\delta), \quad r_A + \frac{b}{c_H} \times (r_H - r_A) < \delta \leq r_H$$

And the optimal solution satisfies:

$$p_A^* = 0, \quad p_H^* = \frac{b - c_R q_R^*}{c_H - c_R q_R^*}, \quad p_R^* = \frac{c_H - b}{c_H - c_R q_R^*}, \quad q_R^* = \frac{(\delta - r_A)c_H + (r_A - r_H)b}{(r_R - r_A)(c_H - b) + (\delta - r_H)c_R}. \quad (5)$$

Unlike the no-collaboration regime (Case 1.3), the available budget is insufficient to rely exclusively on human-only decision making while meeting the performance requirement. At the same time, because the required performance level remains below what humans can achieve on their own, assigning all tasks to AI-assisted review would not induce full engagement from the human. As a result, the coordinator optimally allocates effort across H and R . This regime corresponds to Case 1.2 in Fig. 2a.

As the human budget b increases, reliance on human-only decisions rises and reliance on assistance falls. Moreover, as the performance requirement δ increases, the human's engagement level rises², which increases the share of tasks routed to assisted review. The absence of AI-only decisions reflects that, when $r_H > r_A$ and the system is performance constrained, allocating any mass to A would reduce expected performance without meaningfully relaxing the human effort constraint.

2.2.6 Full Collaboration. In the *full collaboration* regime, all tasks are assigned to assisted decision making. This region exists when the human budget is sufficiently high so that the problem is feasible:

$$\delta > \max\{r_A, r_H\}, \quad b \geq c_R q_R^*.$$

The optimal solution satisfies:

$$p_R^* = 1, \quad p_H^* = 0, \quad p_A^* = 0, \quad q_R^* = \frac{\delta - r_A}{r_R - r_A}. \quad (6)$$

Although this regime features full collaboration between the human and the AI, the human does not fully engage. Instead, the human exerts exactly the minimum level of effort required to satisfy the performance constraint. This effort increases linearly with the stringency of the requirement and decreases with the marginal benefit of review ($r_R - r_A$). This highlights that human engagement is endogenous even under full collaboration, and must be modeled explicitly. This regime corresponds to Case 1.1 (when $r_H > r_A$) and Case 2.1 (when $r_A \geq r_H$) in Fig. 2.

²Take the derivative of the human engagement (5) with respect to δ , we have $\frac{\partial q_R^*}{\partial \delta} = \frac{[c_H(r_R - r_A) + c_R(r_A - r_H)](c_H - b)}{[(r_R - r_A)(c_H - b) + (\delta - r_H)c_R]^2} \geq 0$ since by setting the numerator to be no smaller than 0 gives us $\frac{r_R - r_A}{c_R} \geq \frac{r_H - r_A}{c_H}$, which holds under Assumption 1.

2.3 Discussion of Results and Managerial Implications

Suboptimality of Human–AI Collaboration (Collaboration Paradox). Theorem 2.2 shows that when the performance requirement for a task is modest ($\delta < \max\{r_A, r_H\}$), the optimal coordination policy is to *not* collaborate across a wide range of human budget levels. This holds even though we assume that assisted decision making delivers both higher per-task performance than either human or AI alone ($r_R > \max\{r_H, r_A\}$) and lower per-task cost than human-only processing ($c_R < c_H$). The key driver is that human engagement is endogenous. When reviewing AI-assisted outputs, humans choose their level of effort rather than exerting full scrutiny by default. If AI performance r_A is sufficiently high relative to the required standard δ , routing tasks to assisted review induces only minimal engagement in equilibrium. As a result, collaboration may fail to improve system performance despite favorable per-task characteristics. This mechanism is consistent with field experimental evidence showing that higher-quality algorithmic recommendations can reduce human attention and induce near automatic adherence [Dell’Acqua, 2022, Dell’Acqua et al., 2023].

Phase-Change Behavior (Automation Cliffs). The model predicts sharp changes in optimal coordination as AI performance crosses key thresholds. When AI performance is below human performance and the performance requirement is relatively modest, the optimal policy may involve no collaboration or only partial collaboration (Cases 1.2–1.4), depending on the available human budget. Full collaboration emerges only when performance requirements are sufficiently stringent that neither humans nor AI can meet them alone. This pattern is consistent with many current workflows in which AI has not yet surpassed human performance: unless institutional standards are raised beyond existing human baselines, collaboration is often unnecessary or suboptimal. Once AI performance exceeds human performance, however, these intermediate regimes disappear and are replaced by an AI-only solution. In this sense, the core (single-stage) model exhibits a phase change in optimal coordination driven by an *exogenous* improvement in AI performance.

In practice, organizations are not limited to reacting passively to such exogenous changes. Through mechanisms such as fine-tuning, reinforcement learning with human feedback, and targeted data collection, organizations can *endogenously* improve AI performance within a workflow over time. In Appendix B, we extend the core model to a dynamic setting in which AI performance evolves endogenously through exposure to human expertise. The coordinator’s decisions therefore affect both current task performance and the future capability of the AI system.

3 Conclusion

This paper develops a tractable framework for allocating decision authority between humans and AI under limited oversight capacity and performance requirements. We model the interaction between an institutional coordinator and a human expert as a generalized Nash game in which coordination policies and human engagement are jointly determined. In the static (single-stage) setting, we obtain closed-form equilibrium policies and show that the joint space of performance standards and oversight capacity partitions into a small number of economically interpretable regimes, clarifying when full automation, human-only decisions, or selective human–AI collaboration arise endogenously. Human oversight emerges not simply from technological feasibility but from binding accountability requirements and scarce attention.

Several extensions are promising. First, incorporating richer human behavior—such as heterogeneous expertise, fatigue, or endogenous learning—could refine predictions about oversight allocation and long-run performance. Second, introducing uncertainty about AI quality and model risk would enable the study of robust or risk-sensitive coordination policies. Finally, empirical validation using data from real-world human–AI workflows would help quantify the tradeoffs identified here and inform implementable coordination policies.

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A Proofs for the Core (Single-Stage) Model

This section provides detailed proofs underlying Proposition 2.1, Theorem 2.2, and the regime-wise closed-form equilibria reported in Section 2.2. Throughout, we impose Assumption 1 and the feasibility conditions in Proposition 2.1. We write the coordinator's policy as $p = (p_A, p_H, p_R)$ and the human's engagement as $q \in [0, 1]$ (we suppress the subscript R to simplify notation).

Recall the coordinator's problem:

$$\begin{aligned} \max_{p \geq 0} \quad & U_C(p, q) := r_H p_H + r_A p_A + [r_A + (r_R - r_A)q] p_R \\ \text{s.t.} \quad & c_H p_H + c_R p_R q \leq b, \\ & p_A + p_H + p_R = 1, \end{aligned} \tag{7}$$

and the human's problem:

$$\begin{aligned} \min_{q \in [0, 1]} \quad & U_H(p, q) := c_R p_R q \\ \text{s.t.} \quad & U_C(p, q) \geq \delta. \end{aligned} \tag{8}$$

A.1 Infeasibility and Existence of Equilibrium

A.1.1 Proof for Proposition 2.1. The proof is based on Kakutani's fixed point theorem: Let $S \subset \mathbb{R}^n$ be a nonempty, convex compact set. Let $\Phi : S \rightarrow S$ be a set-valued mapping such that for each $x \in S$, $\Phi(x)$ is a nonempty closed convex subset of S . If Φ is closed on S , then Φ has a fixed point.

We need to verify the conditions of Kakutani's fixed point theorem. The set S for the generalized Nash equilibrium model consists of the feasible domain of the coordinator (7) and the human (8). Under conditions (i) and (ii) of Proposition 2.1, both (7) and (8) are linear programs with nonempty feasible sets, it is clear our set S is nonempty, convex, and compact.

The mapping Φ for our generalized Nash equilibrium model consists of the solution mappings of the coordinator (7) and the human (8). We next show that, for any given point in the set S , the value of the mapping Φ is a nonempty closed convex subset of the set S . Since the linear program (7) has a nonempty bounded constraint set, for any given $q \in [0, 1]$ feasible for the human problem (8), the optimal solution set of the coordinator (7) is nonempty, closed, and convex. Similarly, the linear program (8) has a nonempty bounded constraint set, for any given p feasible for the coordinator's problem (7), the optimal solution set of the human (8) is nonempty, closed, and convex.

It remains to show the mapping Φ for our generalized Nash equilibrium model is closed on set S . To show the solution mappings of the coordinator (7) and the human (8) are closed on their respective domain, we use Proposition 12.1 of [Facchinei and Pang, 2010]. Since the objective functions of the coordinator (7) and the human (8) are continuous, it remains to show that the feasible region for the coordinator (7) is continuous given the human behavior q , and the feasible region for the human (8) is continuous given the coordinator's strategy p . This follows from the well-known fact that a polyhedral graph-convex mapping is Lipschitz continuous on its effective domain (e.g., Example 9.35 of [Rockafellar and Wets, 2009]; Theorem 5.5.8 of [Facchinei and Pang, 2003]). The proof is thus complete.

A.1.2 Infeasible Region. We first formalize the feasibility characterization.

LEMMA A.1 (NECESSARY FEASIBILITY CONDITIONS). *If there exists (p, q) feasible for both (7) and (8), then $\delta \leq r_R$.*

PROOF. Define $\bar{r}_R(q) := r_A + (r_R - r_A)q$. For any p and q , the coordinator's expected reward satisfies

$$U_C(p, q) = r_H p_H + r_A p_A + \bar{r}_R(q) p_R \leq \max\{r_H, r_A, \bar{r}_R(q)\} \leq r_R,$$

where the last inequality uses $\bar{r}_R(q) \leq r_R$ and Assumption 1 (so $r_R \geq r_H, r_A$). Thus $U_C(p, q) \geq \delta$ implies $\delta \leq r_R$. $\square$

Next, consider the strongest feasible policy from the coordinator's perspective: allocate all mass to R . Then the human chooses the minimal q satisfying $U_C(p, q) \geq \delta$.

LEMMA A.2 (MINIMAL ENGAGEMENT UNDER UNIVERSAL ASSISTANCE). *Suppose $p_R = 1$. The minimal q satisfying $U_C(p, q) \geq \delta$ is*

$$q_{\min}(\delta) = \begin{cases} 0, & \delta \leq r_A, \\ \frac{\delta - r_A}{r_R - r_A}, & r_A < \delta \leq r_R, \\ \text{infeasible}, & \delta > r_R. \end{cases}$$

PROOF. If $p_R = 1$, then $U_C(p, q) = \bar{r}_R(q) = r_A + (r_R - r_A)q$, which is increasing in q . Solving $r_A + (r_R - r_A)q \geq \delta$ yields the displayed expression and infeasibility when $\delta > r_R$. $\square$

LEMMA A.3 (SUFFICIENT FEASIBILITY CONDITION). *If $\delta \leq r_R$ and $b \geq c_R q_{\min}(\delta)$, then there exists (p, q) feasible for both (7) and (8).*

PROOF. Choose $p_R = 1$ and $q = q_{\min}(\delta)$. Then by Lemma A.2, $U_C(p, q) \geq \delta$. The coordinator's budget constraint becomes $c_R q_{\min}(\delta) \leq b$, which holds by assumption. Thus (p, q) is jointly feasible. $\square$

Lemmas A.1–A.3 justify the infeasible region discussions in Section 2.2.1: when $\delta > r_R$ or when $\delta > r_A$ but $b < c_R(\delta - r_A)/(r_R - r_A)$, even universal assistance cannot meet the target.

A.1.3 Proof of Theorem 2.2. Existence follows from Proposition 2.1. For characterization, Lemma A.4 shows that the human best response is always the minimal feasible engagement. For any fixed q , the coordinator problem is a linear program and admits an optimal extreme-point policy (Lemma A.6), implying that equilibrium coordination must lie on a finite set of candidate faces, e.g., A only, H only, R only, A – H mix, A – R mix, or H – R mix. For each face, impose the human minimal-feasible response (Lemma A.5) and solve the resulting binding equations to obtain closed-form policies; Propositions A.7–A.11 give the explicit solutions for the regimes that arise under Assumption 1. Finally, the region boundaries are obtained by enforcing feasibility ($q \in [0, 1]$, $p \in \Delta$) and comparing the coordinator's objective across candidate extreme points; these algebraic comparisons yield the partitions depicted in Fig. 2.

A.2 Best Responses of Human and Coordinator

A.2.1 Human Best Response: Minimal Engagement. We now solve the human's optimization given a feasible p .

LEMMA A.4 (HUMAN BEST RESPONSE HAS A THRESHOLD FORM). *Fix p such that the constraint set of (8) is nonempty. Then any human best response $q^*(p)$ is the minimal $q \in [0, 1]$ satisfying $U_C(p, q) \geq \delta$. Moreover:*

- (1) If $p_R = 0$, then $U_C(p, q)$ does not depend on q , and any feasible q minimizes effort; in particular, $q^*(p) = 0$ is optimal.
- (2) If $p_R > 0$, then $U_C(p, q)$ is affine increasing in q with slope $(r_R - r_A)p_R \geq 0$, and hence the minimal feasible q is unique whenever $r_R > r_A$.

PROOF. The human objective is $U_H(p, q) = c_R p_R q$, which is weakly increasing in q . Therefore the human minimizes effort by choosing the smallest feasible q . If $p_R = 0$, then $U_H(p, q) \equiv 0$ and $U_C(p, q)$ does not depend on q , so $q = 0$ is optimal. If $p_R > 0$, then

$$U_C(p, q) = r_H p_H + r_A p_A + [r_A + (r_R - r_A)q] p_R = \underbrace{[r_H p_H + r_A(p_A + p_R)]}_{\text{constant in } q} + (r_R - r_A) p_R q,$$

which is affine increasing in q when $r_R > r_A$. Hence the minimal feasible $q \in [0, 1]$ satisfying $U_C(p, q) \geq \delta$ is unique. $\square$

Explicitly, we can derive closed-form formula for the human (8) as in the lemma below.

LEMMA A.5 (CLOSED-FORM MINIMAL ENGAGEMENT FOR A GIVEN p). *Fix a feasible p with $p_R > 0$ and $r_R > r_A$. Define the baseline reward under the coordinator's policy when the human does not engage:*

$$U_C(p, 0) = r_H p_H + r_A(p_A + p_R).$$

Then the human best response is

$$q^*(p) = \left[\frac{\delta - U_C(p, 0)}{(r_R - r_A)p_R} \right]_0^1, \quad \text{where } [x]_0^1 := \min\{1, \max\{0, x\}\}.$$

PROOF. By Lemma A.4, choose the minimal q such that

$$U_C(p, 0) + (r_R - r_A)p_R q \geq \delta.$$

Solving yields $q \geq (\delta - U_C(p, 0))/((r_R - r_A)p_R)$, and clipping to $[0, 1]$ gives the expression. $\square$

Lemma A.5 shows how the coordinator's choice of (p_A, p_H, p_R) determines the human engagement q needed for existence of an equilibrium.

A.2.2 Coordinator Best Response: Extreme-Point Optimality. Fix $q \in [0, 1]$. The coordinator's problem (7) is a linear program in p . Hence an optimal solution exists at an extreme point of the feasible polytope.

LEMMA A.6 (COORDINATOR OPTIMAL POLICY IS AN EXTREME POINT). *Fix $q \in [0, 1]$ such that the feasible set of (7) is nonempty. Then there exists an optimal solution $p^*(q)$ that is an extreme point of the polytope*

$$\mathcal{P}(q) = \left\{ p \geq 0 : p_A + p_H + p_R = 1, c_H p_H + c_R p_R q \leq b \right\}.$$

PROOF. The objective of the coordinator (7) is linear in p and the constraints define a nonempty compact polytope. By the fundamental theorem of linear programming, e.g., Theorem 2.7 of [Bertsimas and Tsitsiklis, 1997], an optimum is attained at an extreme point. $\square$

A.3 Deriving Each Equilibrium Regime

We now derive the closed-form equilibria by combining (i) the human's minimal feasible response (Appendix A.2.1) and (ii) the coordinator extreme-point optimality (Appendix A.2.2). Because the full partition is summarized in Fig. 2, the goal here is to show how each regime and its policy formulas arise, and to verify equilibrium conditions for each.

A.3.1 *AI-only regime.* This regime appears as Case 2.2 in Fig. 2b.

PROPOSITION A.7 (AI-ONLY EQUILIBRIUM). *Suppose $\delta \leq r_A$ and $r_A \geq r_H$. Then (p^*, q^*) with*

$$p_A^* = 1, p_H^* = p_R^* = 0, q^* = 0$$

is an equilibrium.

PROOF. Since $r_A \geq r_H$, we must have $p_H^* = 0$. Thus, the coordinator's problem (7) reduces to

$$\begin{aligned} \max_{p \geq 0} \quad & r_A p_A + [r_A + (r_R - r_A)q] p_R \\ \text{s.t.} \quad & c_R p_R q \leq b, \\ & p_A + p_R = 1. \end{aligned}$$

Similarly, the human's problem (8) reduces to:

$$\begin{aligned} \min_{q \in [0,1]} \quad & c_R p_R q \\ \text{s.t.} \quad & r_R p_R q + r_A p_R (1 - q) + r_A p_A \geq \delta. \end{aligned}$$

The human can always choose $q = 0$ to make the cost $c_R p_R q = 0$, and at the same time, to satisfy the constraint since $r_R p_R q + r_A p_R (1 - q) + r_A p_A = r_A p_R + r_A p_A = r_A \geq \delta$. Thus, $q^* = 0$ is optimal.

Given $q^* = 0$, the coordinator's problem (7) further becomes

$$\begin{aligned} \max_{p \geq 0} \quad & r_A p_A + r_A p_R \\ \text{s.t.} \quad & 0 \leq b, \\ & p_A + p_R = 1. \end{aligned}$$

In this case, the option R and the option A are equivalent since they yield the same reward r_A when $q^* = 0$. Thus $p_A^* = 1$ and $p_R^* = 0$ is optimal. $\square$

A.3.2 *Human-only regime.* This regime appears as Case 1.4 in Fig. 2a.

PROPOSITION A.8 (HUMAN-ONLY EQUILIBRIUM). *Suppose $\delta \leq r_H$ and $b \geq c_H$, and additionally $r_H > r_A$. Then*

$$p_H^* = 1, p_A^* = p_R^* = 0, q^* = 0$$

is an equilibrium.

PROOF. Since $r_H > r_A$ and $b \geq c_H$, there is no need to use AI, i.e., $p_A^* = 0$. Thus, the coordinator's problem (7) reduces to

$$\begin{aligned} \max_{p \geq 0} \quad & r_H p_H + [r_A + (r_R - r_A)q] p_R \\ \text{s.t.} \quad & c_H p_H + c_R p_R q \leq b, \\ & p_H + p_R = 1. \end{aligned}$$

Similarly, the human's problem (8) reduces to:

$$\begin{aligned} \min_{q \in [0,1]} \quad & c_R p_R q \\ \text{s.t.} \quad & r_R p_R q + r_A p_R (1 - q) + r_H p_H \geq \delta. \end{aligned}$$

Given $p_R = 0$, the human best response is $q^* = 0$ (Lemma A.4). Feasibility holds because $U_C(p, q) = r_R p_R q + r_A p_R (1 - q) + r_H (1 - p_R) = r_H \geq \delta$. Given $q^* = 0$, the coordinator's objective is $U_C(p, q) = r_H p_H + r_A (p_A + p_R)$ and the only cost is $c_H p_H$. Since $b \geq c_H$, the policy $p_H = 1$ is feasible. Moreover, since $r_H > r_A$, to place all mass on H weakly dominates allocating any mass to A (or R with $q = 0$). Thus $p_H^* = 1$ and $p_R^* = 0$ is optimal for the coordinator. $\square$

A.3.3 No-collaboration regime (A-H mix). Here we derive the no-collaboration equilibrium that appears as Case 1.3 in Fig. 2a. In this regime, $p_R^* = 0$, $p_H^*, p_A^* > 0$, and $q^* = 0$.

PROPOSITION A.9 (NO-COLLABORATION EQUILIBRIUM). *Suppose $\delta \leq r_A + b(r_H - r_A)/c_H$, $b < c_H$, and $r_H > r_A$. Then*

$$p_R^* = 0, \quad p_H^* = \frac{b}{c_H}, \quad p_A^* = 1 - \frac{b}{c_H}, \quad q^* = 0$$

is an equilibrium.

PROOF. Fix $q = 0$, the coordinator's problem (7) becomes:

$$\begin{aligned} \max_{p \geq 0} \quad & r_H p_H + r_A p_A + r_R p_R \\ \text{s.t.} \quad & c_H p_H \leq b, \\ & p_A + p_H + p_R = 1. \end{aligned}$$

Because $r_H > r_A$ and the only costly modality is H , the coordinator allocates as much mass to H as the budget permits, i.e., chooses $p_H^* = b/c_H$, with the remainder to A , i.e., $p_A^* = 1 - b/c_H$. Thus $p_R^* = 0$.

Next, we consider the human's best response to p^* . Given p^* , the constraint of performance requirement for human (8) becomes

$$\begin{aligned} r_H p_H^* + r_A p_A^* + [r_A + (r_R - r_A)q] p_R^* &= r_H \times \frac{b}{c_H} + r_A \times \left(1 - \frac{b}{c_H}\right) + [r_A + (r_R - r_A)q] \times 0 \\ &= r_A + (r_H - r_A) \times \frac{b}{c_H} \\ &\geq \delta. \end{aligned}$$

This human constraint holds by assumption, so $q = 0$ is feasible. Given $p_R^* = 0$, by Lemma A.4, $q = 0$ is also optimal for the human. Thus we have $q^* = 0$. $\square$

A.3.4 Full collaboration regime (R only). The full-collaboration equilibrium appears as Case 1.1 in Fig. 2a and Case 1.2 in Fig. 2b. In this regime, $p_R^* = 1$, $p_A^* = p_H^* = 0$, and $q^* > 0$.

PROPOSITION A.10 (FULL COLLABORATION EQUILIBRIUM). *Suppose $\delta > \max\{r_A, r_H\}$, $\delta \leq r_R$, and $b \geq c_R(\delta - r_A)/(r_R - r_A)$. Then*

$$p_R^* = 1, \quad p_A^* = p_H^* = 0, \quad q^* = \frac{\delta - r_A}{r_R - r_A}$$

is an equilibrium.

PROOF. Fix $p_R = 1$. By Lemma A.2, the human best response is $q^* = (\delta - r_A)/(r_R - r_A)$. Consider the coordinator's best response to q^* . Given q^* , the coordinator's budget constraint becomes $c_R q^* \leq b$, which holds by assumption, so $p_R = 1$ is feasible. Moreover, the achieved reward for the assisted decision making (R) is

$$\begin{aligned} r_A + (r_R - r_A)q^* &= r_A + (r_R - r_A) \times \frac{\delta - r_A}{r_R - r_A} \\ &= \delta \\ &> \max\{r_A, r_H\}. \end{aligned}$$

Thus, any deviation that places positive mass on A or H would weakly reduce reward and would not reduce required engagement below q^* in a way that improves the coordinator's objective while maintaining the human constraint at δ . Moreover, back to the perspective of human (8), for fixed

$p_R = 1$ the constraint binds at the minimal-feasible q^* , any reduction in p_R must be compensated by increased q , which will raise the human cost and lead to suboptimal solution for the human. Thus, within the region where $p_R = 1$ is feasible, it is also optimal.

To look at it from a different perspective, since $c_R \leq c_H$, for given q^* , the coordinator's problem (7) reduces to

$$\begin{aligned} \max_{p \geq 0} \quad & \delta p_R \\ \text{s.t.} \quad & c_R q^* \times p_R \leq b, \\ & p_R = 1. \end{aligned}$$

Naturally, $p_R^* = 1$ is the optimal policy for the coordinator. $\square$

A.3.5 Partial collaboration regime (H-R mix). We now derive the partial-collaboration equilibrium that appears as Case 1.2 in Fig. 2a. In this regime, $p_A^* = 0$, $p_H^*, p_R^* > 0$, and $q^* \in (0, 1]$.

PROPOSITION A.11 (PARTIAL COLLABORATION EQUILIBRIUM). Suppose $r_H > r_A$, $r_A + b(r_H - r_A)/c_H < \delta \leq r_H$, and $b \geq c_R(\delta - r_A)/(r_R - r_A)$, then the equilibrium policies are

$$q^* = \frac{(\delta - r_A)c_H + (r_A - r_H)b}{(r_R - r_A)(c_H - b) + (\delta - r_H)c_R}, \quad (9)$$

$$p_R^* = \frac{c_H - b}{c_H - c_R q^*}, \quad p_H^* = \frac{b - c_R q^*}{c_H - c_R q^*}, \quad p_A^* = 0. \quad (10)$$

PROOF. From $r_H > r_A$ and $r_A + b(r_H - r_A)/c_H < r_H$ we have $b < c_H$. Suppose that the equilibrium policy q for the human (8) satisfies $r_A + (r_R - r_A)q \leq r_H$ and $c_R q \leq b$. Given q , according to Lemma A.6, the the best response for the coordinator (7) is the point on the upper convex hull of (c_H, r_H) , $(c_R q, r_A + (r_R - r_A)q)$, and $(0, r_A)$ with maximum value among all convex combinations whose average human budget no larger than b .

Since $r_A \leq r_A + (r_R - r_A)q \leq r_H$ and $c_R q \leq b < c_H$, the best response for the coordinator (7) lines in the convex combination of (c_H, r_H) and $(c_R q, r_A + (r_R - r_A)q)$, subject to the human budget constraint:

$$p_R^* = \frac{c_H - b}{c_H - c_R q}, \quad p_H^* = \frac{b - c_R q}{c_H - c_R q}, \quad p_A^* = 0.$$

It remains to derive the best response q^* of the human (8), and check that $r_A + (r_R - r_A)q^* \leq r_H$ and $c_R q^* \leq b$. Plugging p_R^* , p_H^* , and p_A^* into the human's problem (8),

$$\begin{aligned} \min_{q \in [0,1]} \quad & \frac{c_R(c_H - b)}{c_H/q - c_R} \\ \text{s.t.} \quad & \frac{[(r_R - r_A)(c_H - b) - r_H c_R]q + r_A(c_H - b) + r_H b}{c_H - c_R q} \geq \delta. \end{aligned}$$

Since $c_H - c_R q > 0$ (i.e., $c_H/q - c_R > 0$) and $c_H - b > 0$, the human's problem can be further reformulated as:

$$\begin{aligned} \min_{q \in [0,1]} \quad & q \\ \text{s.t.} \quad & [(r_R - r_A)(c_H - b) + (\delta - r_H)c_R]q \geq (\delta - r_A)c_H + (r_A - r_H)b. \end{aligned}$$

We can show that $(\delta - r_A)c_H + (r_A - r_H)b > 0$:

$$\begin{aligned} & (\delta - r_A)c_H + (r_A - r_H)b > 0 \\ \iff & \delta > r_A + \frac{r_H - r_A}{c_H} b \end{aligned}$$

We can also show that $(r_R - r_A)(c_H - b) + (\delta - r_H)c_R \geq (\delta - r_A)c_H + (r_A - r_H)b$:

$$\begin{aligned} (r_R - r_A)(c_H - b) + (\delta - r_H)c_R &\geq (\delta - r_A)c_H + (r_A - r_H)b \\ \iff \delta &\geq \frac{r_R - r_H}{c_R - c_H}b - \frac{c_H r_R - c_R r_H}{c_R - c_H} \end{aligned}$$

Notice that the point $(b = c_H, \delta = r_H)$ is on both the two lines $\delta = r_A + \frac{b}{c_H}(r_H - r_A)$ and $\delta = \frac{r_R - r_H}{c_R - c_H}b - \frac{c_H r_R - c_R r_H}{c_R - c_H}$. Given $\delta > r_A + \frac{b}{c_H}(r_H - r_A)$, in order to show that the constraint $\delta \geq \frac{r_R - r_H}{c_R - c_H}b - \frac{c_H r_R - c_R r_H}{c_R - c_H}$ is redundant for $b < c_H$, we only need to compare the two slopes $\frac{r_R - r_H}{c_R - c_H}$ and $\frac{r_H - r_A}{c_H}$:

$$\begin{aligned} \frac{r_H - r_A}{c_H} &\leq \frac{r_H - r_A}{c_H} \\ \iff c_H(r_H - r_R) &\leq c_H(r_H - r_A) - c_R(r_H - r_A) \\ \iff c_H(r_A - r_R) &\leq -c_R(r_H - r_A) \\ \iff \frac{r_R - r_A}{c_R} &\geq \frac{r_H - r_A}{c_H} \end{aligned}$$

The last inequality must hold since $r_R \geq r_H$ and $c_R \leq c_H$ under Assumption 1. Moreover, by setting $b = 0$ for the line $\delta = \frac{r_R - r_H}{c_R - c_H}b - \frac{c_H r_R - c_R r_H}{c_R - c_H}$, then we have $\delta = \frac{c_H r_R - c_R r_H}{c_H - c_R} \geq \frac{c_H r_H - c_R r_H}{c_H - c_R} = r_H$.

As a result, $(r_R - r_A)(c_H - b) + (\delta - r_H)c_R \geq (\delta - r_A)c_H + (r_A - r_H)b > 0$, and the human's best response is

$$q^* = \frac{(\delta - r_A)c_H + (r_A - r_H)b}{(r_R - r_A)(c_H - b) + (\delta - r_H)c_R} \in (0, 1]$$

It suffices to check that $r_A + (r_R - r_A)q^* \leq r_H$ and $c_R q^* \leq b$:

(i) $r_A + (r_R - r_A)q^* \leq r_H$

Taking the difference between r_H and $r_A + (r_R - r_A)q^*$, we need to show

$$r_H - [r_A + (r_R - r_A)q^*] = \frac{(r_H - \delta)[(r_R - r_A)c_H - (r_H - r_A)c_R]}{(r_R - r_A)(c_H - b) + (\delta - r_H)c_R} \geq 0$$

Since $(r_R - r_A)(c_H - b) + (\delta - r_H)c_R > 0$, it is equivalent to show that

$$\underbrace{(r_H - \delta)}_{\geq 0} \left[\underbrace{(r_R - r_A)}_{\geq r_H} \underbrace{c_H}_{\geq c_R} - (r_H - r_A)c_R \right] \geq 0$$

(ii) $c_R q^* \leq b$ Subtracting $c_R q^*$ from b , we need to show

$$b - c_R q^* = \frac{(c_H - b)[b(r_R - r_A) - c_R(\delta - r_A)]}{(r_R - r_A)(c_H - b) + (\delta - r_H)c_R} \geq 0$$

Since $c_H > b$ and $(r_R - r_A)(c_H - b) + (\delta - r_H)c_R > 0$, it is equivalent to show that

$$b(r_R - r_A) - c_R(\delta - r_A) \geq 0 \iff b \geq \frac{c_R(\delta - r_A)}{r_R - r_A}$$

This inequality holds since it is the feasibility assumption we have, as in Lemma A.3 and discussed in Section 2.2.1. $\square$

B Extension: Multi-Stage Coordination with Learning

B.1 Problem Setup

The interaction unfolds over a finite horizon of T stages. In each stage $t \in \{0, 1, \dots, T\}$, numerous tasks arrive and are routed by a coordinator to one of three collaboration modalities: full automation (A), human-only decision making (H), or assisted decision making (R). Let $p_t = (p_A^t, p_H^t, p_R^t) \in \Delta^3$ denote the coordinator's (possibly randomized) routing policy in stage t , where $p_A^t + p_H^t + p_R^t = 1$ and $p_A^t, p_H^t, p_R^t \geq 0$. As in the core model, for AI-assisted decision making (R), the AI provides

a recommendation that is shown to the human, and human chooses the level of engagement $q_t \in [0, 1]$ in reviewing the AI output.

A key departure from the core model is that the AI's expected performance is no longer fixed but improves through exposure to human input. We summarize learning with a one-dimensional state variable $x_t \geq 0$, interpreted as cumulative effective human input available before stage t , with $x_0 = 0$. Motivated by statistical learning theory and empirical scaling laws, we assume AI error decays with accumulated supervision and capture this with a simple power-law specification for the AI baseline:

$$r_A^t(x_t) = r_R - \frac{\alpha}{1 + x_t^\beta}, \quad \text{given } \alpha > 0, \beta > 0, \quad (11)$$

Under (11), r_A^t is increasing in x_t and converges to r_R as $x_t \rightarrow \infty$; in particular $r_A^0 = r_R - \alpha$ and α measures the initial gap to the assisted outcome. The parameter β controls the concavity (learning speed). The functional form is a convenient, empirically supported shorthand for the common observation that predictive error falls with additional labeled data, often approximately following a power law [Hestness et al., 2017, Kaplan et al., 2020, Shalev-Shwartz and Ben-David, 2014, Vapnik, 1998]. For transparency, we assume r_R is the asymptotic upper bound on performance attainable with sufficient supervision. This normalization simplifies exposition; none of the qualitative results or the equilibrium phenomena rely on r_R being the exact attainable ceiling.

The learning state evolves as a function of human involvement, with the informativeness of human input depending on current AI quality. While the AI remains below the human baseline, both human-only decisions and human edits of algorithmic recommendations provide useful training signal. Once the AI is beyond human-level performance, further improvement arises primarily from targeted edits produced when humans actively review recommendations. Formally, the state transition is

$$x_{t+1} = x_t + g(x_t; p_t, q_t), \quad g(x_t; p_t, q_t) = \begin{cases} p_H^t + p_R^t q_t, & \text{if } r_A^t \leq r_H, \\ p_R^t q_t, & \text{if } r_A^t > r_H. \end{cases} \quad (12)$$

Given (x_t, p_t, q_t) , the expected stage- t reward equals the weighted performance under the chosen modality:

$$\ell(x_t, p_t, q_t) = r_A^t p_A^t + r_H p_H^t + \left(r_A^t + (r_R - r_A^t) q_t \right) p_R^t. \quad (13)$$

At each stage, the human chooses the engagement level q_t myopically to minimize effort cost subject to meeting a minimum stage-wise performance requirement δ , as in the core (single-stage) model:

$$q_t \in \arg \min_{q \in [0, 1]} c_R p_R^t q \quad \text{s.t.} \quad \ell(x_t, p_t, q) \geq \delta. \quad (14)$$

The human does not internalize the effect of q_t on future AI performance through x_{t+1} .

Anticipating this stage-by-stage best response, the coordinator chooses the routing policy sequence $\{p_t\}_{t=0}^T$ to maximize total reward over the horizon subject to a budget constraint:

$$\begin{aligned} \max_{p_0, \dots, p_T} \quad & \sum_{t=0}^T \ell(x_t, p_t, q_t^*(x_t, p_t)) \\ \text{s.t.} \quad & c_H p_H^t + c_R p_R^t q_t^*(x_t, p_t) \leq b, \quad \forall t \in 0, 1, \dots, T, \\ & p_t \in \Delta^3, \quad \forall t \in 0, 1, \dots, T, \\ & x_{t+1} \text{ follows (12),} \quad x_0 = 0. \end{aligned} \quad (15)$$

where $q_t^*(x_t, p_t)$ denotes the human's optimal response. Closed-form characterizations of q_t^* are provided in Appendix C.1.

B.2 Solution Strategy: Dynamic Programming

Substituting the human best response $q_t^*(x_t, p_t)$ into the coordinator's objective reduces (15) to a single-agent control problem with state x_t and action p_t . Let

$$\tilde{\ell}(x_t, p_t) \triangleq \ell(x_t, p_t, q_t^*(x_t, p_t)), \quad \tilde{g}(x_t, p_t) \triangleq g(x_t, p_t, q_t^*(x_t, p_t)),$$

and define the feasible action set (given state x_t) as

$$\mathcal{P}(x_t) \triangleq \{p_t \in \Delta^3 : c_H p_H^t + c_R p_R^t q_t^*(x_t, p_t) \leq b\}.$$

The value function $V_t(x_t)$ is the maximal continuation payoff from stage t onward starting at state x_t . With terminal condition $V_{T+1}(x_{T+1}) \equiv 0$, the Bellman recursion is

$$V_t(x_t) = \max_{p_t \in \mathcal{P}(x_t)} \left\{ \tilde{\ell}(x_t, p_t) + V_{t+1}(x_t + \tilde{g}(x_t, p_t)) \right\}, \quad t = 0, 1, \dots, T. \quad (16)$$

We solve (16) by backward induction [Bertsekas, 2012, 2017]. This formulation highlights the central intertemporal tradeoff: allocating tasks affects both current performance and the future learning state through its impact on x_t . Implementation details are provided in Appendix D.

A key structural property of the dynamic program is that higher learning states lead to (weakly) better future performance, because the AI baseline improves with accumulated experience. The following lemma formalizes this monotonicity, which will be used repeatedly in characterizing equilibrium regions and transition dynamics.

LEMMA B.1 (MONOTOTICITY OF VALUE FUNCTION). *Suppose $r_A^t(x_t)$ is a monotonically increasing function in x_t , then the value function $V_t(x_t)$ is non-decreasing with respect to x_t .*

B.3 Analysis of Dynamic Policies

Relative to the core (single-stage) model, the key new feature is that coordination decisions influence the evolution of AI performance, so that feasibility and equilibrium policies must be understood jointly with the learning dynamics. Nevertheless, as we show below, the geometric structure of the policy regions remains closely related to the static case: at any given time, the (δ, b) space is partitioned by the same set of boundaries, but these boundaries shift endogenously as the AI improves.

Fig. 5 and 6 illustrate how the equilibrium policies evolve across representative time points. Across all panels, the qualitative regime structure is preserved. What changes over time is the effective AI baseline r_A^t , which increases as the AI learns from human involvement. As r_A^t rises, the region in which AI alone can meet the performance requirement expands, and the portion of the space requiring human-only involvement correspondingly contracts. Thus, the dynamic model can be interpreted as a sequence of static problems with an improving baseline, linked together by the learning process induced by human participation.

These patterns reveal two central mechanisms. First, human involvement generates an intertemporal learning externality: allocating tasks to humans early on improves the AI, which reduces the need for human intervention later. Second, the performance requirement and the human budget jointly determine not only whether oversight is used at a given time but also the speed of the transition toward automation. Stricter performance requirements sustain human involvement for longer horizons, whereas larger budgets accelerate learning and lead to earlier convergence to AI-dominant policies.

We next formalize these observations by characterizing feasibility, establishing equilibrium existence, and analyzing the structure of optimal coordination over time. Detailed proofs are provided in Appendix C.

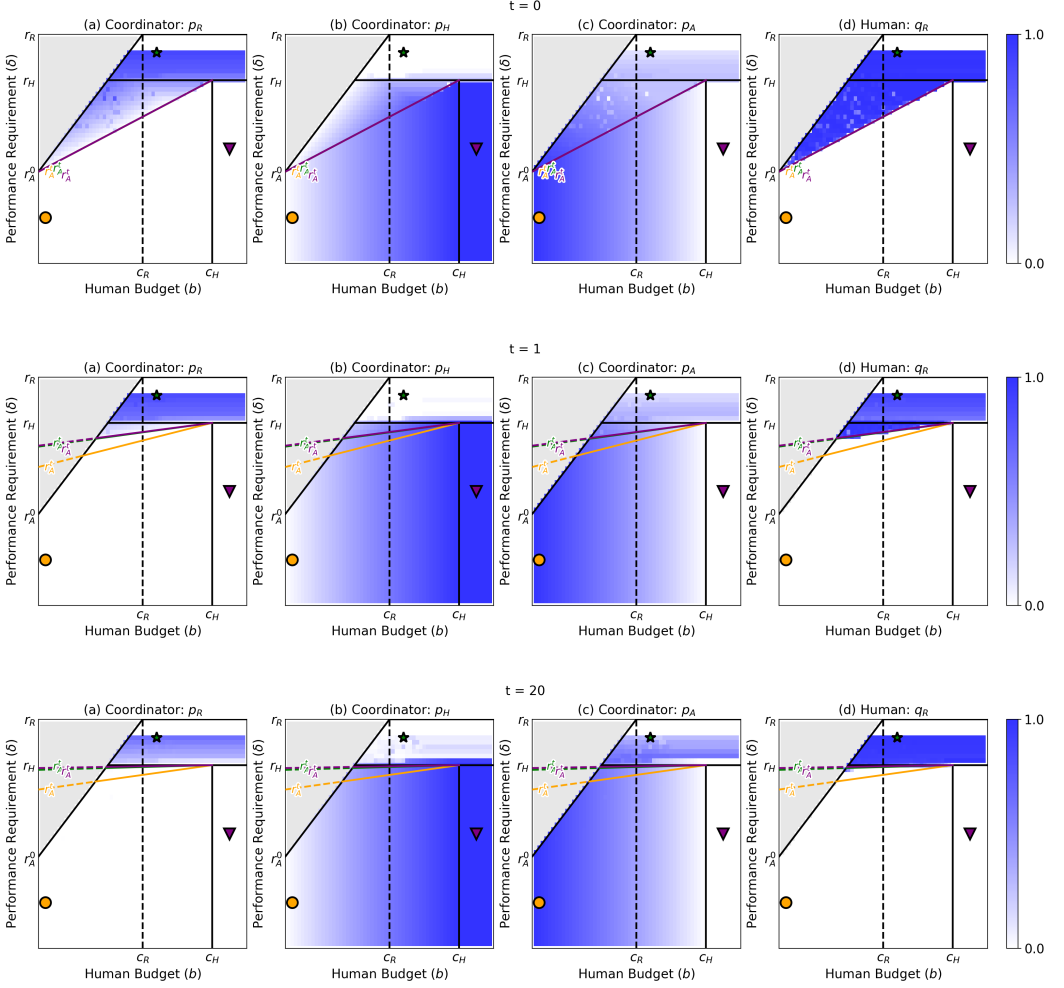


Fig. 5. Evolution of the human and coordinator policies in the early stage of multi-stage model when $t = 0, 1, 20$. Color intensity represents the corresponding policy probability. We illustrate three (δ, b) scenarios along with their associated r_A^t .

B.3.1 Infeasible Region and Existence of Equilibrium. In the multi-stage model, infeasibility is inherently state dependent: for a given budget b and performance requirement δ , feasibility depends on the current AI baseline r_A^t . Nevertheless, the structure of the feasibility constraint is unchanged from the core (single-stage) model. At any fixed state x_t , the feasible region in (b, δ) is the same as in the core (single-stage) model after replacing the constant baseline r_A with the state-dependent baseline $r_A^t(x_t)$. Consequently, under monotone learning (so r_A^t is nondecreasing), the infeasible region can only shrink over time, and the most restrictive (horizon-level) infeasibility is already determined by the initial state x_0 . Lemma B.2 formalizes these statements and gives the explicit boundary characterization.

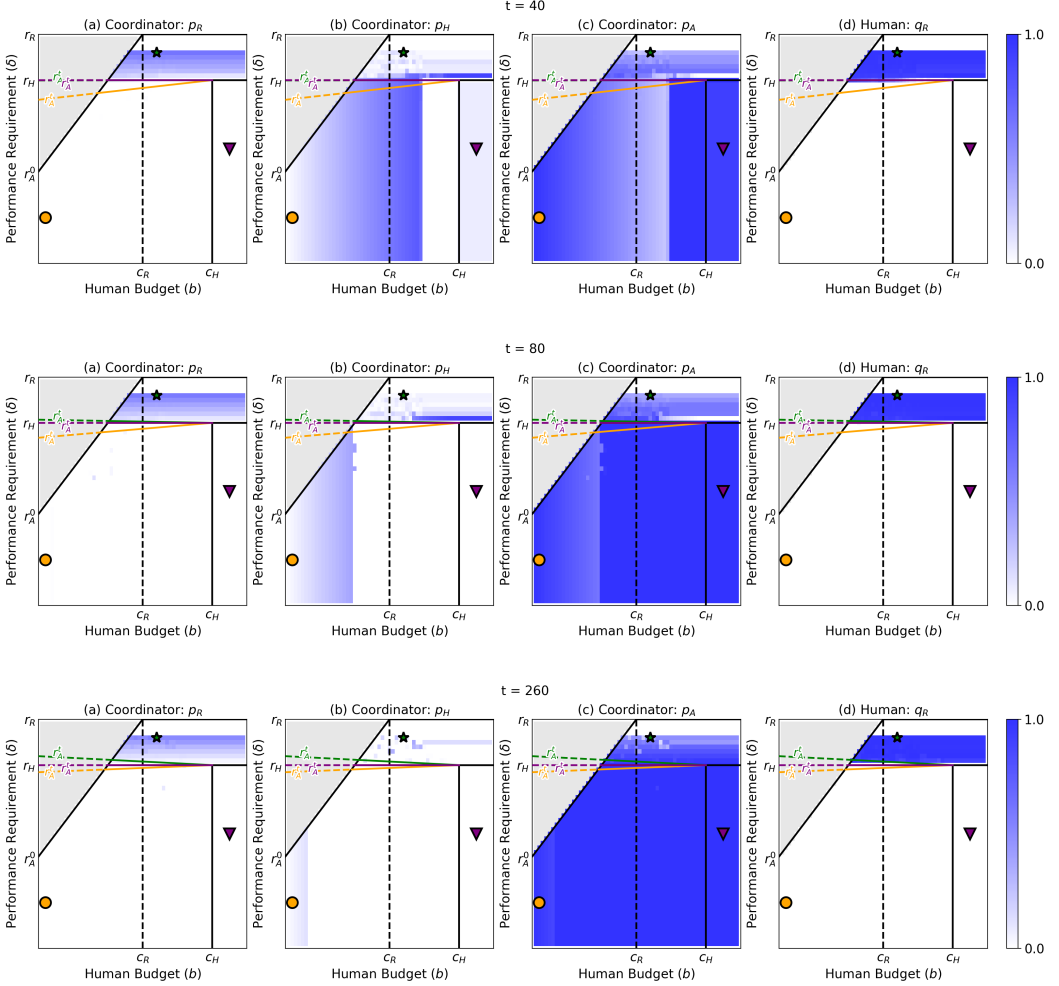


Fig. 6. Evolution of the human and coordinator policies in the later stage when $t = 40, 80, 260$.

LEMMA B.2 (INFEASIBILITY OF THE MULTI-STAGE MODEL). Let $\{x_t\}_{t=0}^T$ be the learning-state sequence induced by a feasible policy in (15). Since r_A^t is nondecreasing in t , the infeasible region of the multi-stage model over the horizon $t = 0, \dots, T$ is

$$\mathcal{I}(\{x_t\}_{t=0}^T) = \left\{ (b, \delta) : \delta > r_R \text{ or } \delta > r_A^0 + \frac{r_R - r_A^0}{c_R} b \right\}.$$

The following proposition shows that whenever the multi-stage problem admits a nonempty feasible set, an equilibrium of the coordinator–human interaction exists.

PROPOSITION B.3 (EQUILIBRIUM CHARACTERIZATION FOR THE MULTI-STAGE MODEL). Suppose the feasibility conditions in Lemma B.2 are satisfied. Then the multi-stage model (15) admits an equilibrium.

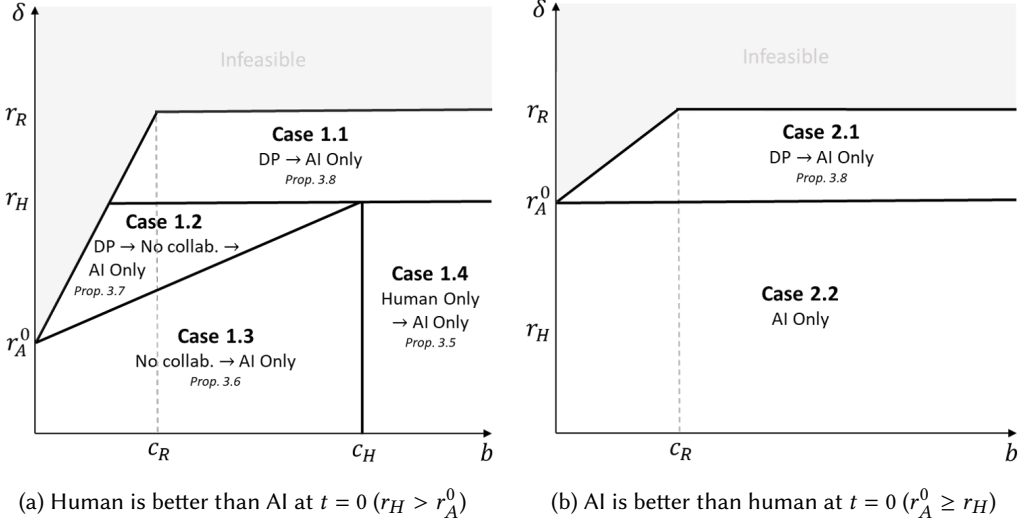


Fig. 7. Equilibrium regions in the multi-stage model.

B.3.2 Regions with Closed-Form Policies. We first identify parameter regions in which the equilibrium policy remains stationary within a phase and changes only when the AI baseline r_A^t crosses a threshold. Fig. 7 summarizes these regions.

LEMMA B.4. *Suppose at certain stage $\hat{t}$ we have $r_A^{\hat{t}} \geq \max\{r_H, \delta\}$, then for stage $t \geq \hat{t}$, the equilibrium policy is*

$$p_A^t = 1, \quad p_H^t = 0, \quad p_R^t = 0, \quad q_t = 0.$$

REMARK 1. *A direct result from Lemma B.4 is that, suppose the feasibility conditions in Lemma B.2 are satisfied, $r_A^0 \geq r_H$, and $\delta \leq r_H$, then the equilibrium policy for the multi-stage model (15) is full automation, i.e., $p_A^t = 1, p_H^t = 0, p_R^t = 0, q_t = 0$ for all $t \in \{0, 1, 2, \dots, T\}$.*

We next consider environments in which AI initially underperforms humans but improves through learning. The key question is how long the system remains human-dependent before transitioning to full automation.

PROPOSITION B.5 (HUMAN-ONLY $\rightarrow$ AI-ONLY). *Suppose $r_A^0 < r_H$, $\delta \leq r_H$, and $b \geq c_H$. The equilibrium policy for the multi-stage model (15) is*

(1) *When $r_A^t < r_H$, i.e., $t < (\frac{\alpha}{r_R - r_H} - 1)^{1/\beta}$,*

$$p_R^t = 0, \quad p_H^t = 1, \quad p_A^t = 0, \quad q_t = 0;$$

(2) *When $r_A^t \geq r_H$, i.e., $t \geq (\frac{\alpha}{r_R - r_H} - 1)^{1/\beta}$,*

$$p_A^t = 1, \quad p_H^t = 0, \quad p_R^t = 0, \quad q_t = 0.$$

Proposition B.5 shows that when sufficient human capacity is available ($b \geq c_H$) and the performance requirement is moderate ($\delta \leq r_H$), the system initially relies entirely on humans. Human decisions both satisfy the performance requirement and accelerate AI learning. Once the AI baseline catches up to the human benchmark, the system transitions endogenously to full automation. The transition time is determined by the learning curve and is independent of the budget.

When the budget is insufficient to sustain full human involvement, the system instead operates in a Non-Collaboration regime prior to Full-Automation.

PROPOSITION B.6 (NO-COLLABORATION $\rightarrow$ AI-ONLY). *Suppose $r_A^0 < r_H$, $\delta \leq r_A^0 + \frac{b}{c_H} \times [r_H - r_A^0]$, and $b < c_H$. The equilibrium policy for the multi-stage model (15) is*

(1) *When $r_A^t < r_H$, i.e., $t < \frac{c_H}{b} (\frac{\alpha}{r_H - r_A^0} - 1)^{1/\beta}$,*

$$p_R^t = 0, \quad p_H^t = \frac{b}{c_H}, \quad p_A^t = 1 - \frac{b}{c_H}, \quad q_t = 0;$$

(2) *When $r_A^t \geq r_H$, i.e., $t \geq \frac{c_H}{b} (\frac{\alpha}{r_H - r_A^0} - 1)^{1/\beta}$,*

$$p_A^t = 1, \quad p_H^t = 0, \quad p_R^t = 0, \quad q_t = 0.$$

Proposition B.6 characterizes the case in which the budget cannot support full human involvement. The system therefore operates in a static mixed regime: a constant fraction b/c_H of decisions is assigned to humans, with the remainder handled by AI. Human involvement gradually improves the AI baseline until it reaches the human benchmark, at which point the system transitions to full automation. Relative to Proposition B.5, the transition occurs later because the effective rate of learning is proportional to the fraction of decisions handled by humans. A tighter budget slows learning and delays automation.

The results above highlight a common structure across regimes. When the AI baseline is below the human benchmark, the system operates in a static phase—either human-only or mixed without collaboration—determined by the budget constraint. During this phase, human involvement drives learning and shifts the AI baseline upward. Once the AI baseline reaches the human benchmark, the system enters an absorbing automation state in which AI alone satisfies the performance requirement. The key implication is that learning induces endogenous regime transitions. Budget and performance requirements determine not only the static allocation of decisions but also the duration of human involvement and the timing of automation. Higher budgets accelerate learning by increasing early human participation, while stricter performance requirements may prolong the period of human reliance.

B.3.3 Regions with Evolving Policies. We now turn to parameter regions in which the equilibrium policy evolves over time as the AI baseline improves. Unlike the closed-form regions in the previous subsection, the system here does not remain in a single stationary regime prior to automation. Instead, the allocation of decisions across AI and human agents adjusts endogenously as learning shifts the effective AI baseline r_A^t .

Two forces jointly shape the transitions. First, when the performance requirement δ is sufficiently high relative to the initial AI baseline, the system may initially require additional coordination or human reliance to meet the requirement. Second, as learning raises r_A^t , the constraint may eventually become slack, allowing the system to enter a simpler mixed regime and ultimately transition to full automation. We characterize these dynamics below.

When the performance requirement is moderately high—above what the initial mixed allocation can satisfy but below the human benchmark—the system exhibits a two-stage transition. It first operates in a regime with active performance constraints and evolving allocations, then transitions into the stationary mixed regime described earlier once learning relaxes the constraint. Finally, when the AI baseline reaches the human benchmark, the system enters the absorbing automation state.

PROPOSITION B.7 (DYNAMIC COORDINATION $\rightarrow$ NO-COLLABORATION $\rightarrow$ AI-ONLY). *Suppose the feasibility conditions in Lemma B.2 are satisfied, $r_A^0 < r_H$, and $r_A^0 + \frac{b}{c_H} \times [r_H - r_A^0] < \delta \leq r_H$. Partial of the equilibrium policy for the multi-stage model (15) can be characterized. Specifically,*

(1) *When $\frac{br_H - c_H \delta}{b - c_H} < r_A^t < r_H$, the equilibrium policy is*

$$p_R^t = 0, \quad p_H^t = \frac{b}{c_H}, \quad p_A^t = 1 - \frac{b}{c_H}, \quad q_t = 0;$$

(2) *When $r_A^t \geq r_H$, the equilibrium policy is*

$$p_A^t = 1, \quad p_H^t = 0, \quad p_R^t = 0, \quad q_t = 0.$$

Proposition B.7 shows that when the performance requirement initially exceeds what the mixed allocation can satisfy, the system must rely more heavily on human involvement early on. As learning raises the AI baseline, the performance constraint eventually becomes slack, allowing the system to operate in the stationary mixed regime characterized in Proposition B.6. From that point onward, learning continues at a constant effective rate until the AI baseline reaches the human benchmark, triggering the transition to full automation. Relative to the closed-form regions in the previous subsection, this regime exhibits an additional intermediate phase driven by the performance requirement. Stricter requirements therefore delay the arrival of the stationary mixed regime and, in turn, delay automation.

We next consider environments in which the performance requirement exceeds the human benchmark r_H . In such cases, automation cannot occur until the AI baseline itself satisfies the requirement.

PROPOSITION B.8 (DYNAMIC COORDINATION $\rightarrow$ AI-ONLY). *Suppose the feasibility conditions in Lemma B.2 are satisfied, $\delta \geq r_H$, and $r_A^0 \leq \delta$. When $r_A^t \geq \delta$, the equilibrium policy for the multi-stage model (15) is*

$$p_A^t = 1, \quad p_H^t = 0, \quad p_R^t = 0, \quad q_t = 0.$$

Proposition B.8 shows that when the performance requirement exceeds the human benchmark, even fully relying on humans is insufficient for long-run feasibility. Automation becomes viable only once AI performance itself crosses the required threshold. Until that point, human involvement serves primarily as a learning mechanism rather than a permanent solution. The timing of automation is therefore governed directly by the learning curve and the required performance level: stricter requirements push the transition further into the future.

B.3.4 Analysis of the Learning Dynamics. We now analyze the learning dynamics of the multi-stage model in (15). Our goal is to understand whether the expected performance of the AI system (r_A^t) can eventually reach the performance level of human experts (r_H) or the required system performance threshold (δ) within a finite number of stages.

We first note that under Cases 1.3 or 1.4 in Fig. 7, the expected performance of AI always reaches the level of human performance in finite time. In particular, there exists a finite stage $t < \infty$ such that $r_A^t \geq r_H \geq \delta$. The exact time at which r_A^t reaches r_H is characterized in Proposition B.5 and Proposition B.6. Therefore, the main remaining question concerns the reachability of Cases 1.1 and 2.1 in Fig. 7.

To analyze the evolution of AI performance, consider stage $t \in \{0, 1, \dots, T-1\}$ and fix the cumulative effective human input x_t . Define the AI learning rate at stage t as the change in expected AI performance between stages t and $t+1$:

$$\Delta r_A^t(x_t, p_t) \triangleq r_A^{t+1}(x_{t+1}) - r_A^t(x_t) = \frac{\alpha}{1 + x_t^\beta} - \frac{\alpha}{1 + [x_t + \tilde{g}(x_t, p_t)]^\beta}.$$

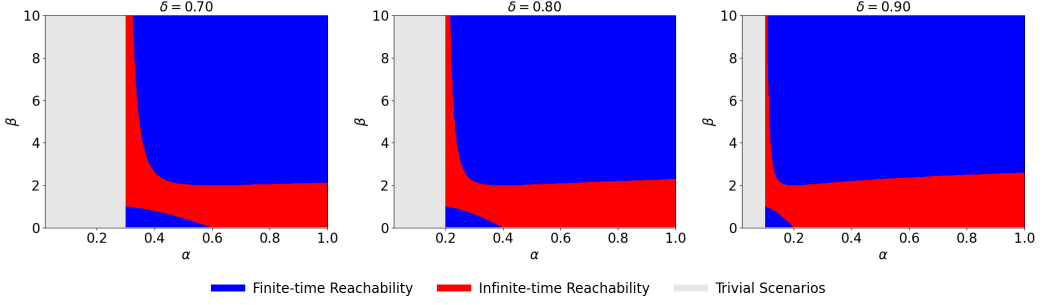


Fig. 8. Empirical analysis of learning dynamics under the multi-stage model when $r_R = 1$ at the slowest learning rate (when $p_H^t = 0$ is fixed).

From this expression we observe that larger values of $\tilde{g}(x_t, p_t)$ lead to a larger learning increment $\Delta r_A^t(x_t, p_t)$, and therefore faster improvement in AI performance. The next lemma shows that human-only decisions accelerate AI learning by establishing that the learning rate increases with the probability of human-only decisions p_H^t . In particular, $p_H^t = 0$ produces the *slowest learning rate*, while $p_H^t = 1$ yields the *fastest learning rate*.

LEMMA B.9. *Suppose the feasibility conditions in Lemma B.2 are satisfied. At stage $t \in \{0, 1, \dots, T-1\}$ and for a fixed x_t , if $\hat{p}_H^t \geq p_H^t$, then*

$$\Delta r_A^t(x_t, \hat{p}_t) \geq \Delta r_A^t(x_t, p_t), \quad \forall \hat{p}_t, p_t \in \Delta^3.$$

We first study the slowest learning scenario. Proposition B.10 characterizes whether AI performance can reach the required system threshold δ when the learning rate is minimal (i.e., $p_H^t = 0$). Under conditions (1) or (2), δ is reached in finite time; under conditions (3)–(5), AI performance approaches δ only asymptotically as $T \rightarrow \infty$.

PROPOSITION B.10 (REACHABILITY OF PERFORMANCE REQUIREMENT WHEN $p_H^t = 0$). *Suppose the feasibility conditions in Lemma B.2 are satisfied, $\delta > r_H$, and $r_A^0 \leq \delta$. Assume that the learning rate of AI is the slowest (i.e., $p_H^t = 0$). For the multi-stage model (15), there always exists stage $\hat{t} < \infty$ such that $r_A^{\hat{t}} \geq \delta$ under one of the following conditions:*

- (1) $\beta > 1$ and $\left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} + \left(\frac{\alpha}{r_R - \delta} - 1\right)^{(1-\beta)/\beta} < \beta$;
- (2) $0 < \beta < 1$ and $\left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} + \left(\frac{\alpha}{r_R - \delta} - 1\right)^{(1-\beta)/\beta} \leq 1$.

In general, such a finite $\hat{t}$ that $r_A^{\hat{t}} \geq \delta$ may not exist. Under one of the following conditions, we have $r_A^t < \delta$ for all $t \in \{0, 1, \dots, T\}$:

- (3) $\beta > 1$ and $\left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} + \left(\frac{\alpha}{r_R - \delta} - 1\right)^{(1-\beta)/\beta} \geq \beta$;
- (4) $\beta = 1$;
- (5) $0 < \beta < 1$ and $\left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} + \left(\frac{\alpha}{r_R - \delta} - 1\right)^{(1-\beta)/\beta} > 1$.

Corollary B.11 further characterizes the asymptotic limit of r_A^T as the time horizon grows.

COROLLARY B.11. *Suppose the feasibility conditions in Lemma B.2 are satisfied, $\delta > r_H$, and $r_A^0 \leq \delta$. Assume the slowest learning rate ($p_H^t = 0$). For the multi-stage model (15):*

(1) Under condition (1) of Proposition B.10,

$$\delta \leq \lim_{T \rightarrow \infty} r_A^T \leq r_R - \frac{\alpha}{1 + \left(\left(\frac{\alpha}{r_R - \delta} - 1 \right)^{1/\beta} + 1 - \frac{r_R - \delta}{\alpha} \right)^\beta}.$$

(2) Under condition (2),

$$r_A^0 = r_R - \alpha, \quad r_A^t = r_R - \frac{\alpha}{1 + \left(1 - \frac{r_R - \delta}{\alpha} \right)^\beta}, \quad \forall t \geq 1.$$

(3) Under conditions (3)–(5),

$$\lim_{T \rightarrow \infty} r_A^T = \delta.$$

Fig. 8 visualizes the regions characterized in Proposition B.10. The blue region corresponds to parameter values under which r_A^t reaches δ in finite time, whereas the red region represents parameter values where convergence to δ occurs only asymptotically. Two observations are worth noting. First, when β is small, the AI performance tends to approach δ only asymptotically. This occurs because β controls the speed of performance improvement in (11): smaller values of β lead to slower learning when the cumulative human input x_t is large. Second, when both α and β are small, r_A^t may reach δ quickly. In this case the initial AI performance $r_A^0 = r_R - \alpha$ is already close to δ . This result is numerically consistent with (11). When the cumulative effective human input is smaller than 1, a smaller β leads to faster AI learning, which is the opposite of the behavior observed when x_t becomes large. However, this scenario is largely pathological in practice, since such a small level of cumulative human input is already sufficient for the AI performance r_A^t to reach the required threshold δ . Lastly, the gray region in each subplot corresponds to the trivial case $r_A^0 > \delta$, where the performance requirement is satisfied at $t = 0$.

We next consider the fastest learning scenario. The following proposition shows that when human-only decisions occur with probability one ($p_H^t = 1$), the AI system always reaches the performance threshold δ in finite time regardless of the parameter values.

PROPOSITION B.12 (REACHABILITY OF PERFORMANCE REQUIREMENT WHEN $p_H^t = 1$). *Suppose the feasibility conditions in Lemma B.2 are satisfied, $\delta > r_H$, and $r_A^0 \leq \delta$. Assume the fastest learning rate ($p_H^t = 1$). Then for the multi-stage model (15), there exists a finite stage $\hat{t} < \infty$ such that $r_A^{\hat{t}} \geq \delta$. Moreover,*

$$\lim_{T \rightarrow \infty} r_A^T = r_R.$$

Finally, we analyze the reachability of Case 1.2 in Fig. 7. The argument proceeds in two steps. First, even under the slowest learning rate ($p_H^t = 0$), the AI performance eventually exceeds the threshold $(br_H - c_H\delta)/(b - c_H)$. Second, once $r_A^t > (br_H - c_H\delta)/(b - c_H)$, Proposition B.7 implies that the state transition reduces to $x_{t+1} = x_t + \frac{b}{c_H}$ when $r_A^t \leq r_H$. Since the increment $b/c_H > 0$, the cumulative effective human input x_t increases monotonically. Consequently, the AI performance defined in (11) eventually satisfies $r_A^t \geq r_H \geq \delta$ for some finite stage t .

B.4 Discussion of Results and Managerial Implications

Consistency between core and extended model. The multi-stage model preserves the core managerial implications from the static analysis. When the initial AI baseline r_A^0 lies below the human benchmark r_H and the performance requirement is modest, the dynamic optimum may never use assisted review and the system transitions directly to AI-only once r_A^t becomes sufficiently high. Likewise, whether the initial AI baseline exceeds or falls short of r_H continues to induce the same

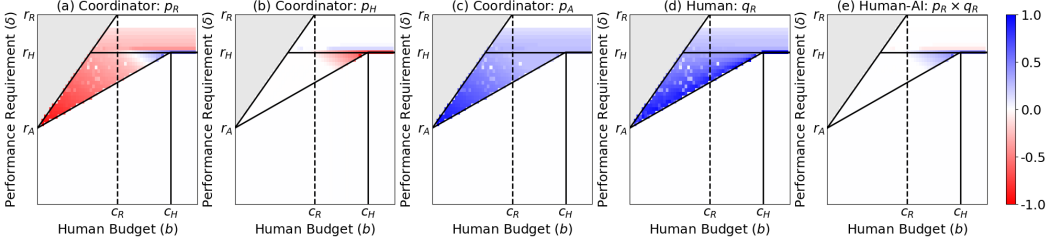


Fig. 9. Difference between the initial policy ($t = 0$) in the multi-stage model and the static policy in the core (single-stage) model.

phase-change behavior as in the core (single-stage) model: the relative ranking of r_A^0 and r_H remains a first-order determinant of coordination regimes.

Differences Introduced by Learning. The dynamic objective introduces a forward value for human involvement, which alters early allocations. Fig. 9 compares the initial policy in the multi-stage model to the core (single-stage) model and exposes a consistent pattern. Panels (a)–(c) show that the shares of tasks assigned to direct human processing (p_H) and to assisted routing (p_R) are typically smaller in the optimal multi-stage policy than in the optimal core policy. This reduction in task volume is offset by higher per-task effort: panel (d) shows a higher engagement probability q_R on routed tasks, and panel (e) shows that the product $p_R q_R$, i.e. the effective reviewed fraction, rises over substantial regions. Thus the optimal multi-stage policy concentrates human input: fewer tasks receive human attention, but those that do receive more intensive review.

The mechanism is strategic: the coordinator reallocates scarce human effort toward tasks where marginal learning value is largest (typically when the AI baseline is near the human benchmark) so as to both satisfy short-run requirements and accelerate long-run improvement. In consequence, total human effort in the early stages can increase relative to the static benchmark even while the fraction of tasks routed to humans falls. Put differently, learning induces a shift from broad, shallow human involvement to narrow, deep interventions that maximize the informational return on review effort.

C Proofs for the Multi-Stage Model

C.1 Closed-form Solution for Human’s Optimal Response in the Multi-Stage Model

In this appendix, we derive the closed-form solution for the human’s problem (14) at each stage $t = \{0, 1, 2, \dots, T\}$. From the constraint $\ell(x_t, p_t, q_t) \geq \delta$,

$$\begin{aligned} \ell(x_t, p_t, q_t) &= r_A^t p_A^t + r_H p_H^t + \left(r_A^t + (r_R - r_A^t) q_t \right) p_R^t \\ &= (r_R - r_A^t) p_R^t q_t + r_A^t p_A^t + r_H p_H^t + r_A^t p_R^t \\ &\geq \delta \end{aligned}$$

Since $r_R - r_A^t > 0$ by definition, we have

$$p_R^t q_t \geq \frac{\delta - r_A^t p_A^t - r_H p_H^t - r_A^t p_R^t}{r_R - r_A^t}$$

Recall that we also have $q_t \in [0, 1]$. For given policy, p_t , of the coordinator in stage t , the human (14) aims to minimize the effort cost, $c_R p_R^t q_t$. There are four possible situations,

(i) When $p_R^t = 0$:

If $\delta - r_A^t p_A^t - r_H p_H^t \leq 0$, then we have $q_t^*(x_t, p_t) = 0$.

If $\delta - r_A^t p_A^t - r_H p_H^t > 0$, the problem is infeasible.

(ii) When $p_R^t \neq 0$:

$$q_t \geq \frac{\delta - r_A^t p_A^t - r_H p_H^t - r_A^t p_R^t}{[r_R - r_A^t] p_R^t}$$

If $\delta - r_A^t p_A^t - r_H p_H^t - r_A^t p_R^t < 0$, we have $q_t^*(x_t, p_t) = 0$.

If the right-hand side of this inequality is within $[0, 1]$, i.e., $p_H^t r_H + (1 - p_H^t) r_A^t \leq \delta \leq p_R^t r_R + p_H^t r_H + p_A^t r_A^t$, we have

$$\begin{aligned} q_t^*(x_t, p_t) &= \frac{\delta - r_A^t p_A^t - r_H p_H^t - r_A^t p_R^t}{[r_R - r_A^t] p_R^t} \\ &= \frac{\delta - r_A^t [p_A^t + p_R^t] - r_H p_H^t}{[r_R - r_A^t] p_R^t} \\ &= \frac{\delta - r_A^t + [r_A^t - r_H] p_H^t}{[r_R - r_A^t] p_R^t} \end{aligned}$$

If the right-hand side of this inequality is larger than 1, i.e., $\delta > p_R^t r_R + p_H^t r_H + p_A^t r_A^t$, the problem is infeasible.

Summing up the analysis above, we have

$$q_t^*(x_t, p_t) = \begin{cases} \frac{\delta - r_A^t + [r_A^t - r_H] p_H^t}{[r_R - r_A^t] p_R^t}, & \text{if } p_R^t \neq 0, \delta \geq p_H^t r_H + (1 - p_H^t) r_A^t, \\ & \text{and } \delta \leq p_R^t r_R + p_H^t r_H + p_A^t r_A^t \\ 0, & \text{if } p_R^t \neq 0 \text{ and } \delta < p_H^t r_H + (1 - p_H^t) r_A^t \\ 0, & \text{if } p_R^t = 0 \text{ and } \delta \leq p_H^t r_H + (1 - p_H^t) r_A^t \\ \text{infeasible}, & \text{if } p_R^t \neq 0 \text{ and } \delta > p_R^t r_R + p_H^t r_H + p_A^t r_A^t \\ \text{infeasible}, & \text{if } p_R^t = 0 \text{ and } \delta > p_H^t r_H + (1 - p_H^t) r_A^t \end{cases}$$

C.2 Monotonicity of Value Function, Infeasibility, and Existence of Equilibrium

C.2.1 Proof for Proposition B.1: Monotonicity of Value Function. We prove this lemma by induction. By definition, $V_{T+1}(x_{T+1}) \equiv 0$ is non-decreasing with respect to (w.r.t.) x_{T+1} . First, we show $V_T(x_T) = \max_{p_T \in \mathcal{P}(x_T)} \tilde{\ell}(x_T, p_T)$ is non-decreasing w.r.t. x_T . As defined in Section B.2, $V_T(x_T)$ can be explicitly written as

$$\begin{aligned} \max_{p_T \geq 0} \quad & \ell(x_T, p_T, q_T^*) := r_A^T p_A^T + r_H p_H^T + \left(r_A^T + (r_R - r_A^T) q_T^* \right) p_R^T \\ \text{s.t.} \quad & c_H p_H^T + c_R p_R^T q_T^* \leq b, \\ & p_A^T + p_H^T + p_R^T = 1, \end{aligned}$$

where q_T^* is the optimal policy of the human at stage T defined as

$$\begin{aligned} q_T^* &\in \operatorname{argmin}_{q \in [0, 1]} c_R p_R^T q \\ \text{s.t.} \quad & \ell(x_T, p_T, q) \geq \delta. \end{aligned}$$

Since function $r_A^T(x_T)$ is monotonically increasing in x_T , for given pair (p_T, q) , the function $\ell(x_T, p_T, q)$ is also monotonically increasing in x_T . Thus, as x_T increases, the set of optimal solution for the human will not become smaller since the constraint is easier to satisfy, namely the human

will not have fewer optimal policy q_T^* to choose from. From the coordinator's perspective, as x_T increases, the objective function $\ell(x_T, p_T, q_T^*)$ increases and the constraint set will not become smaller (since there are possibly more q_T^* for the human to choose from). As a result, $V_T(x_T)$ is non-decreasing w.r.t. x_T .

Next, suppose $V_{t+1}(x_{t+1})$ is non-decreasing w.r.t. x_{t+1} , we show $V_t(x_t)$ is non-decreasing w.r.t. x_t . From the Bellman equation (16),

$$V_t(x_t) = \max_{p_t \in \mathcal{P}(x_t)} \left\{ \tilde{\ell}(x_t, p_t) + V_{t+1}(x_t + \tilde{g}(x_t, p_t)) \right\}.$$

To write it down explicitly,

$$\begin{aligned} V_t(x_t) = \max_{p_t \geq 0} & \left\{ \ell(x_t, p_t, q_t^*) + V_{t+1}(x_t + \tilde{g}(x_t, p_t)) \right\} \\ \text{s.t.} & \quad c_H p_H^t + c_R p_R^t q_t^* \leq b, \\ & \quad p_A^t + p_H^t + p_R^t = 1, \end{aligned}$$

where q_t^* is the optimal policy of the human at stage t defined as

$$\begin{aligned} q_t^* \in \operatorname{argmin}_{q \in [0,1]} & \quad c_R p_R^t q \\ \text{s.t.} & \quad \ell(x_t, p_t, q) \geq \delta. \end{aligned}$$

We can have a similar argument as to show that $V_T(x_T) = \max_{p_T \in \mathcal{P}(x_T)} \tilde{\ell}(x_T, p_T)$ is non-decreasing w.r.t. x_T . Because the functions $r_A^t(x_t)$ and $\ell(x_t, p_t, q)$ are monotonically increasing w.r.t. x_t , as x_t increases, the optimal policy set $\{q_t^*\}$ for the human will not be smaller. Furthermore, since the function $V_{t+1}(x_{t+1})$ is non-decreasing w.r.t. x_{t+1} , the function $V_{t+1}(x_t + \tilde{g}(x_t, p_t))$ is also non-decreasing w.r.t. x_t . As a result, for the coordinator, the objective function will increase and the constraint set will not be smaller. Consequently, $V_t(x_t)$ is non-decreasing w.r.t. x_t .

C.2.2 Proof for Lemma B.2: Infeasible Region. Fix a learning state x_t . Define the state-dependent infeasible set

$$\mathcal{I}(x_t) \triangleq \left\{ (b, \delta) : \nexists (p_t, q_t) \in \Delta^3 \times [0, 1] \text{ s.t. } c_H p_H^t + c_R p_R^t q_t \leq b \text{ and } \ell(x_t, p_t, q_t) \geq \delta \right\},$$

where ℓ is given in (13). This is the same notion as the single-stage infeasible region in Section 2.2.1, with the AI baseline r_A replaced by r_A^t .

For a fixed x_t , the following boundaries characterize the infeasible region in (b, δ) space.

- (1) Performance upper boundary: Because $\ell(x_t, p_t, q_t) \leq r_R$ for all (p_t, q_t) , every point with

$$\delta > r_R$$

is infeasible. Thus the upper boundary is the horizontal line $\delta = r_R$.

- (2) Universal-assistance (sufficient) boundary: For $\delta \in (r_A^t, r_R]$, consider universal assistance $p_R = 1$. The minimal modification intensity needed to meet δ is

$$q_{\min}(x_t, \delta) = \frac{\delta - r_A^t}{r_R - r_A^t},$$

so universal assistance is feasible if

$$b \geq c_R q_{\min}^t(x_t, \delta) = \frac{c_R}{r_R - r_A^t} (\delta - r_A^t) \Leftrightarrow \delta \leq \frac{r_R - r_A^t}{c_R} b + r_A^t.$$

Summarizing (i) and (ii), for fixed learning rate x_t , the infeasible set is defined as follows

$$\mathcal{I}(x_t) \triangleq \left\{ (b, \delta) : \delta > r_R \text{ and } \delta > \frac{r_R - r_A^t}{c_R} b + r_A^t \right\}.$$

Let $x_0 = 0$ and given the coordinator's policy $\{p_t\}_{t=0}^T$, then all the possible states, $\{x_t\}_{t=0}^T$, for the multi-stage model (15) can be computed from (12). Then the infeasible set for each x_t is $\mathcal{I}(x_t)$. All the lines with respect to the (b, δ) pair,

$$\delta = \frac{r_R - r_A^t}{c_R} b + r_A^t,$$

intersects at the point $(b = c_R, \delta = r_R)$. Moreover, for any monotonically increasing function r_A^t , e.g., the one defined in (11), the slope $[r_R - r_A^t]/c_R$ decreases with respect to x_t . As a result,

$$\mathcal{I}(x_T) \subseteq \mathcal{I}(x_{T-1}) \subseteq \cdots \subseteq \mathcal{I}(x_2) \subseteq \mathcal{I}(x_1) \subseteq \mathcal{I}(x_0).$$

Taking the intersection for the infeasible sets of all possible states $\{x_t\}_{t=0}^T$, the infeasible set of the multi-stage model (15) is

$$\mathcal{I}(\{x_t\}_{t=0}^T) \triangleq \bigcap_{t=0}^T \mathcal{I}(x_t) = \mathcal{I}(x_0) = \left\{ (b, \delta) : \delta > r_R \text{ and } \delta > \frac{r_R - r_A^0}{c_R} b + r_A^0 \right\}.$$

C.2.3 Proof for Proposition B.3: Existence of The Multi-stage Equilibrium. Since the feasibility conditions in Lemma B.2 are satisfied, the constraint set of the multi-stage model (15) is nonempty. Recall that the objective function of (15) is to maximize

$$L(\{p_t\}_{t=0}^T) = \sum_{t=0}^T \ell(x_t, p_t, q_t^*(x_t, p_t)).$$

In order to show that the single-agent control problem (15) admits a solution, we use Weierstrass' Theorem (e.g., Proposition 2.1.1 of [Bertsekas et al., 2003]): Consider a closed proper function $-L : \{P(x_t)\}_{t=0}^T \rightarrow (-\infty, \infty]$, if its effective domain, defined as

$$\text{dom}(-L) \triangleq \left\{ (p_t \in \mathcal{P}(x_t))_{t=0}^T : -L(\{p_t\}_{t=0}^T) < \infty \right\},$$

is bounded, then the set of minima of $-L$ over the feasible set $\{P(x_t)\}_{t=0}^T$ is nonempty and compact.

We need to verify the conditions of Weierstrass' Theorem. First of all, $\text{dom}(-L)$ is bounded and $-L$ is a proper function by definition,

$$\begin{aligned} -L(\{p_t\}_{t=0}^T) &= -\sum_{t=0}^T \ell(x_t, p_t, q_t^*(x_t, p_t)) \\ &\geq -(T+1)r_R > -\infty, \end{aligned}$$

and

$$\begin{aligned} -L(\{p_t\}_{t=0}^T) &= -\sum_{t=0}^T \ell(x_t, p_t, q_t^*(x_t, p_t)) \\ &\leq -(T+1) \min\{r_H, r_A^0\} < \infty. \end{aligned}$$

It remains to show $-L$ is a closed function. According to Proposition 1.2.3 of [Bertsekas et al., 2003], it suffices to show that (i) L is upper semicontinuous (u.s.c.) over $\text{dom}(L)$; (ii) $\text{dom}(-L)$ is closed:

Step 1: We show $(x_t, p_t) \mapsto g(x_t, p_t)$ is u.s.c. for each $t \in \{0, 1, \dots, T\}$.

First, we show $(x_t, p_t) \mapsto p_R^t q_t^*(x_t, p_t)$ is continuous for each $t \in \{0, 1, \dots, T\}$. Since the feasibility conditions in Lemma B.2 are satisfied, we only need to consider the feasible domain. The closed-form solution for the human (14) at stage $t \in \{0, 1, \dots, T\}$, as derived in Appendix C.1, can be written as

$$q_t^*(x_t, p_t) = \begin{cases} \frac{\delta - r_A^t + [r_A^t - r_H] p_H^t}{[r_R - r_A^t] p_R^t}, & \text{if } p_R^t \neq 0, \delta \geq p_H^t r_H + (1 - p_H^t) r_A^t, \\ & \text{and } \delta \leq p_R^t r_R + p_H^t r_H + p_A^t r_A^t \\ 0, & \text{if } p_R^t \neq 0 \text{ and } \delta < p_H^t r_H + (1 - p_H^t) r_A^t \\ 0, & \text{if } p_R^t = 0 \text{ and } \delta \leq p_H^t r_H + (1 - p_H^t) r_A^t \end{cases}$$

Furthermore, we have

$$p_R^t q_t^*(x_t, p_t) = \left[\frac{\delta - r_A^t + (r_A^t - r_H) p_H^t}{r_R - r_A^t} \right]_+, \quad (17)$$

where $[z]_+ := \max\{z, 0\}$. The denominator in (17) is strictly positive since $r_R - r_A^t > 0$. Also, r_A^t is continuous in x_t by definition. Hence $(x_t, p_t) \mapsto p_R^t q_t^*(x_t, p_t)$ is continuous.

Next, we show $(x_t, p_t) \mapsto g(x_t, p_t)$ is u.s.c. for each $t \in \{0, 1, \dots, T\}$. Define set $S_t = \{x_t \geq 0 : r_A^t \leq r_H\}$, which is an open set since r_A^t is continuous. Using (17), the state increment in (12) can be written as

$$g(x_t; p_t) = p_R^t q_t^*(x_t, p_t) + p_H^t \mathbf{1}_{S_t}(x_t), \quad (18)$$

where $\mathbf{1}_{S_t}(x_t)$ is the indicator function of S_t , namely $\mathbf{1}_{S_t}(x_t) = 1$ if $x_t \in S_t$, and $\mathbf{1}_{S_t}(x_t) = 0$ otherwise. Since the mapping $(x_t, p_t) \mapsto p_R^t q_t^*(x_t, p_t)$ is continuous, and the indicator function $x_t \mapsto \mathbf{1}_{S_t}(x_t)$ is u.s.c. in closed set S_t (e.g., Chapter 1.B of [Rockafellar and Wets, 2009]). Therefore, as their composition, $(x_t, p_t) \mapsto g(x_t, p_t)$ is u.s.c.

Step 2: Denote $\mathbf{p} \triangleq (p_0, p_1, \dots, p_T)$. We show that for each $t \in \{0, 1, \dots, T\}$, $\mathbf{p} \mapsto x_t(\mathbf{p})$ is u.s.c.

Recall that the state transition (12) is

$$x_0 = 0, \quad x_{t+1} = x_t + g(x_t; p_t).$$

Take any sequence $\mathbf{p}^{(n)} \mapsto \mathbf{p}$. By definition of u.s.c., we need to show

$$x_t(\mathbf{p}) \geq \limsup_{n \rightarrow \infty} x_t(\mathbf{p}^{(n)}), \quad \forall t \in \{0, 1, \dots, T\}.$$

We prove this by induction. When $t = 0$, it holds since $x_0 \equiv 0$. Suppose the claim holds for t . Consider x_{t+1} :

$$x_{t+1}(\mathbf{p}^{(n)}) = x_t(\mathbf{p}^{(n)}) + g(x_t(\mathbf{p}^{(n)}); p_t^{(n)}).$$

Taking lim sup on both sides, and using subadditivity of lim sup, i.e., $\limsup (a_n + b_n) \leq \limsup a_n + \limsup b_n$, we have

$$\limsup_n x_{t+1}(\mathbf{p}^{(n)}) \leq \limsup_n x_t(\mathbf{p}^{(n)}) + \limsup_n g(x_t(\mathbf{p}^{(n)}); p_t^{(n)}).$$

By the induction hypothesis and u.s.c. of function g ,

$$\limsup_n x_{t+1}(\mathbf{p}^{(n)}) \leq x_t(\mathbf{p}) + g(x_t(\mathbf{p}); p_t) = x_{t+1}(\mathbf{p}),$$

which is exactly the u.s.c. of $x_{t+1}(\mathbf{p})$. Thus the claim also holds for $t + 1$.

Step 3: We show $\mathbf{p} \mapsto \tilde{\ell}(x_t(\mathbf{p}), p_t)$ is u.s.c. for each $t \in \{0, 1, \dots, T\}$. Consequently, $\mathbf{p} \mapsto L(\mathbf{p})$ is also u.s.c.

Let $\tilde{\ell}(x_t, p_t) \triangleq \ell(x_t, p_t, q_t^*(x_t, p_t))$. From (17) we have,

$$\tilde{\ell}(x_t, p_t) = \begin{cases} \delta, & \text{if } \delta \geq p_H^t r_H + (1 - p_H^t) r_A^t, \\ p_H^t r_H + (1 - p_H^t) r_A^t, & \text{otherwise.} \end{cases}$$

Hence $\tilde{\ell}(x_t, p_t)$ is continuous and non-decreasing in x_t . Furthermore, since $x_t(\mathbf{p})$ is u.s.c., the composition $\mathbf{p} \mapsto \tilde{\ell}(x_t(\mathbf{p}), p_t)$ is u.s.c. (e.g., Exercise 1.40 of [Rockafellar and Wets, 2009]). Therefore, the function

$$L(\mathbf{p}) = \sum_{t=0}^T \tilde{\ell}(x_t(\mathbf{p}), p_t)$$

is also u.s.c as a finite sum of u.s.c. functions.

Step 4: It remains to show that the effective domain of function $-L$, $\text{dom}(-L)$, is closed.

Recall that for state x_t and action $p_t = (p_A^t, p_H^t, p_R^t) \in \Delta^3$, the human budget constraint is

$$c_H p_H^t + c_R p_R^t q_t^*(x_t, p_t) \leq b$$

With (17), the human budget constraint can be further written as

$$c_H p_H^t + c_R \left[\frac{\delta - r_A^t + (r_A^t - r_H) p_H^t}{r_R - r_A^t} \right]_+ \leq b$$

Define

$$B(x_t, p_t) := c_H p_H^t + c_R \left[\frac{\delta - r_A^t + (r_A^t - r_H) p_H^t}{r_R - r_A^t} \right]_+$$

$B(x_t, p_t)$ is continuous in (x_t, p_t) . Moreover, we can show that $B(x_t, p_t)$ is non-increasing in r_A^t , and hence non-increasing in x_t . From the feasibility conditions in Lemma B.2, we have $\delta \leq p_H r_H + (1 - p_H) r_R$. This monotonicity can be verified by differentiating the interior (per- $[\cdot]_+$) term w.r.t. $a := r_A^t$:

$$\frac{d}{da} \left[\frac{\delta - a + (a - r_H) p_H}{r_R - a} \right] = \frac{\delta - (p_H r_H + (1 - p_H) r_R)}{(r_R - a)^2} \leq 0.$$

Now take $\mathbf{p}^{(n)} \rightarrow \mathbf{p}$ with each $\mathbf{p}^{(n)}$ is feasible, i.e.,

$$B(x_t(\mathbf{p}^{(n)}), p_t^{(n)}) \leq b \quad \forall t, n.$$

From Step 2, we know that for each $t \in \{0, 1, \dots, T\}$, $\mathbf{p} \mapsto x_t(\mathbf{p})$ is u.s.c.. Since $B(x_t, p_t)$ is continuous and non-increasing in x_t , and $\mathbf{p} \mapsto x_t(\mathbf{p})$ is u.s.c., the composition $\mathbf{p}^{(n)} \rightarrow B(x_t(\mathbf{p}^{(n)}), p_t^{(n)})$ is lower semicontinuous (l.s.c.) (e.g., Exercise 1.40 of [Rockafellar and Wets, 2009]). Thus, by definition of l.s.c.,

$$B(x_t(\mathbf{p}), p_t) \leq \liminf_{n \rightarrow \infty} B(x_t(\mathbf{p}^{(n)}), p_t^{(n)}) \leq b.$$

Therefore, the human budget constraint is preserved under limits, and the set of feasible policy sequences $\mathbf{p}^{(n)}$ is closed. Consequently, the effective domain of function $-L$, $\text{dom}(-L)$, is closed.

C.3 Proofs for Region with Dynamic Policies

C.3.1 Proof of Lemma B.4. Since $r_A^t \geq r_H$ for all stage $t \geq \hat{t}$, from the coordinator's perspective, we have $p_H^t = 0$ for stage $t \geq \hat{t}$. Thus, the human's optimal policy at stage $t \geq \hat{t}$ in Appendix C.1 reduces to

$$q_t^*(x_t, p_t) = \begin{cases} \frac{\delta - r_A^t}{[r_R - r_A^t] p_R^t}, & \text{if } p_R^t \neq 0, r_A^t \leq \delta \leq p_R^t r_R + (1 - p_R^t) r_A^t, \\ 0, & \text{if } p_R^t \neq 0 \text{ and } \delta < r_A^t, \\ 0, & \text{if } p_R^t = 0 \text{ and } \delta \leq r_A^t \end{cases}$$

Since $r_A^t \geq \delta$ for all stage $t \geq \hat{t}$, we must have $q_t^*(x_t, p_t) = 0$ for stage $t \geq \hat{t}$.

Given $q_t^*(x_t, p_t) = 0$ for stage $t \geq \hat{t}$, next we show that $p_H^t = 0$ for stage $t \geq \hat{t}$ is indeed the optimal for the coordinator. From the coordinator's perspective, the expected stage- t reward (13) reduces to

$$\hat{\ell}(x_t, p_t, q_t^*(x_t, p_t)) = r_A^t p_A^t + r_A^t p_R^t + r_H p_H^t, \quad \forall t \in \{\hat{t}, \hat{t} + 1, \dots, T\},$$

and the per-stage budget constraint for human effort becomes:

$$c_H p_H^t \leq b, \quad \forall t \in \{\hat{t}, \hat{t} + 1, \dots, T\}.$$

Then the multi-stage problem (15) becomes

$$\begin{aligned} & \max_{\{p_t\}_{t=0}^T} \sum_{t=0}^{\hat{t}-1} \ell(x_t, p_t, q_t^*(x_t, p_t)) + \sum_{t=\hat{t}}^T (r_A^t p_A^t + r_A^t p_R^t + r_H p_H^t) \\ & \text{s.t. } c_H p_H^t + c_R p_R^t q_t^*(x_t, p_t) \leq b, \quad \forall t \in \{0, 1, \dots, \hat{t} - 1\}, \\ & \quad c_H p_H^t \leq b, \quad \forall t \in \{\hat{t}, \hat{t} + 1, \dots, T\} \\ & \quad p_t \in \Delta^3, \quad \forall t \in \{0, 1, \dots, \hat{t} - 1\}, \\ & \quad x_{t+1} \text{ follows (12)}. \end{aligned}$$

For $t \geq \hat{t}$, the option R and the option A are equivalent since they yield the same reward r_A^t and the same per-stage budget. Moreover, the option H is not as good as the option R or the option A since $r_A^t \geq r_H$ and the option H has a cost c_H . Thus $p_H^t = p_R^t = 0$ and $p_A^t = 1$ is the equilibrium policy for the multi-stage model (15) at stage $t \geq \hat{t}$.

C.3.2 Proof of Proposition B.5: Human-Only $\rightarrow$ AI-Only. From the Bellman recursion (16) of the multi-stage model (15), at each stage $t \in \{0, 1, \dots, T\}$,

$$V_t(x_t) = \max_{p_t \in \mathcal{P}(x_t)} \left\{ \tilde{\ell}(x_t, p_t) + V_{t+1}(x_t + \tilde{g}(x_t, p_t)) \right\}. \quad (19)$$

Based on the closed-form solution of the human, $q_t^*(x_t, p_t)$, as in Appendix C.1, the function $\tilde{g}(x_t, p_t)$ in the state transition (12) can be computed as

$$\tilde{g}(x_t, p_t) = g(x_t, p_t, q_t^*(x_t, p_t)) = \begin{cases} p_H^t + p_R^t q_t^*(x_t, p_t) = p_H^t + \left[\frac{\delta - r_A^t + (r_A^t - r_H) p_H^t}{r_R - r_A^t} \right]_+, & \text{if } r_A^t \leq r_H, \\ p_R^t q_t^*(x_t, p_t) = \left[\frac{\delta - r_A^t + (r_A^t - r_H) p_H^t}{r_R - r_A^t} \right]_+, & \text{if } r_A^t > r_H. \end{cases}$$

where $[z]_+ := \max\{z, 0\}$. We can observe that with larger p_H^t (i.e., more direct human involvement), $\tilde{g}(x_t, p_t)$ will not decrease. By Lemma B.1, we know that $V_{t+1}(x_t + \tilde{g}(x_t, p_t))$ is non-decreasing in $x_t + \tilde{g}(x_t, p_t)$. Thus, for fixed x_t , $V_{t+1}(x_t + \tilde{g}(x_t, p_t))$ is non-decreasing in p_H^t .

When $r_A^t < r_H$, $\delta \leq r_H$, and $b \geq c_H$, for the first part of (19), from Theorem 2.2 and the description in Section 2.2.3, the equilibrium policy for $\max_{p_t \in \mathcal{P}(x_t)} \tilde{\ell}(x_t, p_t)$ is $p_H^t = 1$ and $p_R^t = p_A^t = q_t = 0$. Since $V_{t+1}(x_t + \tilde{g}(x_t, p_t))$ is non-decreasing in p_H^t for fixed x_t , this equilibrium policy also simultaneously maximizes $V_{t+1}(x_t + \tilde{g}(x_t, p_t))$, the second part of (19). As a result, this equilibrium policy maximizes V_t , and it is the equilibrium policy for the multi-stage model (15) at stage t . Under this equilibrium policy, let $r_A^t = r_H$ using (11), then we have $t = (\frac{\alpha}{r_R - r_H} - 1)^{1/\beta}$.

When $r_A^t \geq r_H$, $\delta \leq r_H$, and $b \geq c_H$, from Lemma B.4, the equilibrium policy for the multi-stage model (15) is $p_A^t = 1$ and $p_H^t = p_R^t = q_t = 0$.

C.3.3 Proof of Proposition B.6: No-Collaboration $\rightarrow$ AI-Only. When $r_A^t < r_H$, the proof is similar to that of Proposition B.5. We also need to analyze (19), the Bellman equation of the multi-stage model (15) at each stage $t \in \{0, 1, \dots, T\}$. The main difference compared to the proof of Proposition B.5 is that, when analyzing the first part of (19), the equilibrium policy for $\max_{p_t \in \mathcal{P}(x_t)} \tilde{\ell}(x_t, p_t)$ is $p_H^t = \frac{b}{c_H}$, $p_A^t = 1 - \frac{b}{c_H}$, and $p_R^t = q_t = 0$, according to Theorem 2.2 and the description in Section 2.2.4. Since in this case $b < c_H$, the largest p_H^t possible is $p_H^t = \frac{b}{c_H}$. Thus, this equilibrium policy also maximizes $V_{t+1}(x_t + \tilde{g}(x_t, p_t))$, and consequently, maximizes $V_t(x_t)$ as their summation. Under this equilibrium policy, let $r_A^t = r_H$ using (11), then we have $t = (\frac{\alpha}{\frac{c_H}{b}(\frac{\alpha}{r_R - r_H} - 1)} - 1)^{1/\beta}$. When $r_A^t \geq r_H$, the result can be derived from Lemma B.4. The proof is thus completed.

C.3.4 Proof of Proposition B.7: Dynamic Coordination $\rightarrow$ No-Collaboration $\rightarrow$ AI-Only. When $r_A^t \geq r_H$, the result can be derived from Lemma B.4. It remains to prove for the case when $\frac{br_H - c_H\delta}{b - c_H} < r_A^t < r_H$. Fix $p_R^t = 0$, $p_H^t = \frac{b}{c_H}$, and $p_A^t = 1 - \frac{b}{c_H}$, then the human's optimal policy at stage $t \geq \hat{t}$ in Appendix C.1 reduces to $q_t^*(x_t, p_t) = 0$.

Assume at certain stages t_1 and t_2 , we have $r_A^{t_1} = \frac{br_H - c_H\delta}{b - c_H}$ and $r_A^{t_2} = r_H$, respectively. Next, we consider the coordinator's best response to $q_t^*(x_t, p_t) = 0$. Given $q_t^*(x_t, p_t) = 0$ for stage $t_1 < t < t_2$, from the coordinator's perspective, the per-stage budget constraint for human effort becomes:

$$c_H p_H \leq b, \quad \forall t \in \{t_1 + 1, t_1 + 2, \dots, t_2 - 1\}.$$

Thus, $p_H^t = \frac{b}{c_H}$ is feasible for stage $t \in \{t_1 + 1, t_1 + 2, \dots, t_2 - 1\}$.

To show $p_H^t = \frac{b}{c_H}$ is also optimal, we take a look at the the Bellman recursion (16) of the multi-stage model (15), at each stage $t \in \{t_1 + 1, t_1 + 2, \dots, t_2 - 1\}$,

$$V_t(x_t) = \max_{p_t \in \mathcal{P}(x_t)} \left\{ \tilde{\ell}(x_t, p_t) + V_{t+1}(x_t + \tilde{g}(x_t, p_t)) \right\}.$$

From $r_A^0 < r_H$ and $r_A^0 + \frac{b}{c_H} \times [r_H - r_A^0] < \delta \leq r_H$, we have $b < c_H$. Furthermore, based on $\frac{br_H - c_H\delta}{b - c_H} < r_A^t$, we have $r_A^t + \frac{b}{c_H} \times [r_H - r_A^t] < \delta$. Similar to the proof of Proposition B.6, for the first part of the per-stage Bellman equation, $\max_{p_t \in \mathcal{P}(x_t)} \tilde{\ell}(x_t, p_t)$, the equilibrium policy is $p_H^t = \frac{b}{c_H}$, $p_A^t = 1 - \frac{b}{c_H}$, and $p_R^t = 0$, from Theorem 2.2 and the description in Section 2.2.4. At the same time, $p_H^t = \frac{b}{c_H}$ is also the largest value possible since in this case $b < c_H$. Recall that in Lemma B.1 we have shown that $V_{t+1}(x_t + \tilde{g}(x_t, p_t))$ is non-decreasing in $x_t + \tilde{g}(x_t, p_t)$, and in the proof of Proposition B.5 we have shown that $\tilde{g}(x_t, p_t)$ is non-decreasing in p_H . Thus, this equilibrium policy also maximizes $V_{t+1}(x_t + \tilde{g}(x_t, p_t))$, the second part of the per-stage Bellman equation. As a result, this equilibrium policy maximizes $V_t(x_t)$ as their summation, and $p_H^t = \frac{b}{c_H}$ is also optimal.

C.3.5 Proof of Proposition B.8: Dynamic Coordination $\rightarrow$ AI-Only. The condition here, $r_A^t \geq \delta \geq r_H$, satisfies the condition in Lemma B.4 by setting $\hat{t} = 0$. Based on Lemma B.4, we can derive the equilibrium policy for the multi-stage model (15) as listed.

C.3.6 Proof of Lemma B.9. We only need to consider the feasible region in Lemma B.2. Based on the closed-form solution of the human, $q_t^*(x_t, p_t)$, as in Appendix C.1, the function $\tilde{g}(x_t, p_t)$ in the state transition (12) can be computed as

$$\tilde{g}(x_t, p_t) = g(x_t, p_t, q_t^*(x_t, p_t)) = \begin{cases} p_H^t + p_R^t q_t^*(x_t, p_t) = p_H^t + \left[\frac{\delta - r_A^t + (r_A^t - r_H)p_H^t}{r_R - r_A^t} \right]_+, & \text{if } r_A^t \leq r_H, \\ p_R^t q_t^*(x_t, p_t) = \left[\frac{\delta - r_A^t + (r_A^t - r_H)p_H^t}{r_R - r_A^t} \right]_+, & \text{if } r_A^t > r_H. \end{cases}$$

where $[z]_+ := \max\{z, 0\}$. We can observe that with larger p_H^t (i.e., more direct human involvement), $\tilde{g}(x_t, p_t)$ will not decrease:

(i) When $r_A^t > r_H$,

$$\tilde{g}(x_t, p_t) = \left[\frac{\delta - r_A^t + (r_A^t - r_H)p_H^t}{r_R - r_A^t} \right]_+.$$

It is trivial that the function $\tilde{g}(x_t, p_t)$ is non-decreasing in p_H^t since $r_R > r_A^t$.

(ii) When $r_A^t \leq r_H$ and $\delta - r_A^t + (r_A^t - r_H)p_H^t \leq 0$, $\tilde{g}(x_t, p_t) = p_H^t$, which is a strictly increasing function with respect to p_H^t .

(iii) When $r_A^t \leq r_H$ and $\delta - r_A^t + (r_A^t - r_H)p_H^t > 0$, we have

$$\tilde{g}(x_t, p_t) = p_H^t + \frac{\delta - r_A^t + (r_A^t - r_H)p_H^t}{r_R - r_A^t} = \frac{\delta - r_A^t + (r_R - r_H)p_H^t}{r_R - r_A^t}.$$

Since $r_R > r_H$ and $r_R > r_A^t$, the function $\tilde{g}(x_t, p_t)$ is a strictly increasing function with respect to p_H^t .

C.3.7 Proof of Proposition B.10 and Corollary B.11. According to Lemma B.9, the slowest AI learning rate gives us $p_H^t = 0$ for all $t \in \{0, 1, \dots, T\}$. When $r_A^t = \delta$, from (11) we have $x_t = \left(\frac{\alpha}{r_R - \delta} - 1 \right)^{1/\beta}$.

Let $\hat{x} \triangleq \left(\frac{\alpha}{r_R - \delta} - 1 \right)^{1/\beta} > 0$.

For any $x_t \geq 0$,

$$\frac{\delta - r_A^t}{r_R - r_A^t} = \frac{\delta - \left(r_R - \frac{\alpha}{1+x_t^\beta} \right)}{\frac{\alpha}{1+x_t^\beta}} = 1 - \frac{(r_R - \delta)(1+x_t^\beta)}{\alpha}.$$

Using $1 + \hat{x}^\beta = \alpha / (r_R - \delta)$, we obtain that

$$\frac{\delta - r_A^t}{r_R - r_A^t} = 1 - \frac{1 + x_t^\beta}{1 + \hat{x}^\beta} = \frac{\hat{x}^\beta - x_t^\beta}{1 + \hat{x}^\beta}.$$

Thus, the state transition (12) becomes

$$x_{t+1} = x_t + \max\left\{ \frac{\delta - r_A^t}{r_R - r_A^t}, 0 \right\} = x_t + \max\left\{ \frac{\hat{x}^\beta - x_t^\beta}{1 + \hat{x}^\beta}, 0 \right\}, \quad \forall t = 0, 1, 2, \dots, T-1. \quad (20)$$

Assume $x_t < \hat{x}$. Then $x_t^\beta < \hat{x}^\beta$, hence the increment in (20) is positive and

$$x_{t+1} > x_t. \quad (21)$$

Also, $x_t = \hat{x}$ implies $x_{t+1} = \hat{x}$. Thus, starting from $x_0 = 0 \leq \hat{x}$, the sequence is nondecreasing and one of the following must happen: (i) either $x_t \geq \hat{x}$ for some finite t (finite-time crossing), or (ii)

$x_t < \hat{x}$ for all t and $x_t \uparrow \hat{x}$ (approach from below without ever reaching). So the question reduces to: *under what conditions does each of these two situations happen?*

Fix any $x \in [0, \hat{x}]$ and define the gap after one update:

$$\hat{x} - x_{t+1} = \hat{x} - x_t - \frac{\hat{x}^\beta - x_t^\beta}{1 + \hat{x}^\beta}.$$

For notational clarity in this step only, write $x = x_t$. Then

$$\hat{x} - x_{t+1} = (\hat{x} - x) - \frac{\hat{x}^\beta - x^\beta}{1 + \hat{x}^\beta}. \quad (22)$$

Crossing occurs at time $t + 1$ if and only if $x_{t+1} \geq \hat{x}$, i.e., $\hat{x} - x_{t+1} \leq 0$.

When $\beta > 1$. Consider the function on $[0, \hat{x}]$: $f(x) \triangleq \hat{x} - x - \frac{\hat{x}^\beta - x^\beta}{1 + \hat{x}^\beta}$. Then we have $f(\hat{x}) = 0$. Differentiate it:

$$f'(x) = -1 + \frac{\beta x^{\beta-1}}{1 + \hat{x}^\beta}.$$

Since $\beta > 1$, the map $x \mapsto x^{\beta-1}$ is increasing on $[0, \hat{x}]$, hence $f'(x)$ is increasing on $[0, \hat{x}]$. In particular,

$$f'(x) \leq f'(\hat{x}) \quad \forall x \in [0, \hat{x}].$$

(1) If $f'(\hat{x}) \leq 0$, then there is no finite-time crossing. If $f'(\hat{x}) \leq 0$, then $f'(x) \leq 0$ for all $x \in [0, \hat{x}]$, so f is nonincreasing on $[0, \hat{x}]$. Since $f(\hat{x}) = 0$ and f is nonincreasing, we get

$$f(x) \geq 0 \quad \forall x \in [0, \hat{x}].$$

Therefore, whenever $x_t \in [0, \hat{x}]$,

$$\hat{x} - x_{t+1} = f(x_t) \geq 0 \quad \Longleftrightarrow \quad x_{t+1} \leq \hat{x}.$$

Starting from $x_0 = 0 \in [0, \hat{x}]$, induction gives $x_t \leq \hat{x}$ for all t . Moreover, if $x_t < \hat{x}$, from (21) then we have $x_{t+1} > x_t$. Thus, the sequence is strictly increasing and bounded above by $\hat{x}$, hence converges to some limit $\leq \hat{x}$ (by monotone convergence theorem). The only possible limit is $\hat{x}$ because $\hat{x}$ is the unique fixed point of (20) in $[0, \infty)$ (since $x_t \mapsto r_A^t(x_t)$ is strictly increasing), so

$$x_t \uparrow \hat{x} \quad \text{and} \quad x_t < \hat{x} \quad \forall t,$$

i.e., no finite $\hat{t}$ with $x_{\hat{t}} \geq \hat{x}$ exists and $\lim_{T \rightarrow \infty} x_t = \hat{x}$, i.e., $\lim_{T \rightarrow \infty} r_A^T = \delta$. The condition $f'(\hat{x}) = -1 + \frac{\beta \hat{x}^{\beta-1}}{1 + \hat{x}^\beta} \leq 0$ can be written explicitly as:

$$\left(\frac{\alpha}{r_R - \delta} - 1 \right)^{1/\beta} + \left(\frac{\alpha}{r_R - \delta} - 1 \right)^{(1-\beta)/\beta} \geq \beta$$

(2) If $f'(\hat{x}) > 0$, then finite-time crossing occurs. If $f'(\hat{x}) > 0$, then by continuity of f' there exists $\varepsilon > 0$ such that

$$f'(x) > 0 \quad \forall x \in (\hat{x} - \varepsilon, \hat{x}].$$

Since $f(\hat{x}) = 0$ and f is strictly increasing on $(\hat{x} - \varepsilon, \hat{x}]$, it follows that for all $x \in (\hat{x} - \varepsilon, \hat{x})$,

$$f(x) < f(\hat{x}) = 0.$$

Thus, whenever $x_t \in (\hat{x} - \varepsilon, \hat{x})$,

$$\hat{x} - x_{t+1} = f(x_t) < 0 \quad \Longleftrightarrow \quad x_{t+1} > \hat{x},$$

so the next iterate crosses $\hat{x}$. Finally, because x_t is increasing and bounded above by $\hat{x}$ until it crosses, we have $x_t \uparrow \hat{x}$ as long as it stays below $\hat{x}$, hence it must eventually enter the interval

$(\hat{x} - \varepsilon, \hat{x})$, triggering a crossing at the next step. Therefore, a finite $\hat{t}$ with $x_{\hat{t}} \geq \hat{x}$ exists. From (20) we have

$$x_{t+1} = x_t + \max\left\{\frac{\hat{x}^\beta - x_t^\beta}{1 + \hat{x}^\beta}, 0\right\} \leq x_t + \frac{\hat{x}^\beta}{1 + \hat{x}^\beta}.$$

Thus,

$$\hat{x} \leq \lim_{T \rightarrow \infty} x_T \leq \hat{x} + \frac{\hat{x}^\beta}{1 + \hat{x}^\beta}.$$

From (11), we have

$$\delta \leq \lim_{T \rightarrow \infty} r_A^T \leq r_R - \frac{\alpha}{1 + \left(\left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} + 1 - \frac{r_R - \delta}{\alpha}\right)^\beta}.$$

Furthermore, the condition $f'(\hat{x}) = -1 + \frac{\beta \hat{x}^{\beta-1}}{1 + \hat{x}^\beta} > 0$ can be written explicitly as:

$$\left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} + \left(\frac{\alpha}{r_R - \delta} - 1\right)^{(1-\beta)/\beta} < \beta$$

When $\beta = 1$. If $\beta = 1$, for $x_t \in [0, \hat{x}]$ (20) becomes

$$x_{t+1} = x_t + \frac{\hat{x} - x_t}{1 + \hat{x}} = \frac{\hat{x}}{1 + \hat{x}} x_t + \frac{\hat{x}}{1 + \hat{x}}.$$

Solve the linear recursion:

$$x_T = \hat{x} \left(1 - \left(\frac{\hat{x}}{1 + \hat{x}}\right)^T\right).$$

Since $\hat{x} > 0$, then $0 < \hat{x}/(1 + \hat{x}) < 1$ so $x_t < \hat{x}$ for every finite t , and $\lim_{T \rightarrow \infty} x_T \rightarrow \hat{x}$, i.e., $\lim_{T \rightarrow \infty} r_A^T = \delta$.

When $0 < \beta < 1$. If $0 < \beta < 1$, then x^β is concave on $[0, \infty)$. For any $x \in [0, \hat{x}]$, concavity implies

$$x^\beta \geq \frac{x}{\hat{x}} \hat{x}^\beta + \left(1 - \frac{x}{\hat{x}}\right) 0 = \frac{x}{\hat{x}} \hat{x}^\beta,$$

Thus,

$$\hat{x}^\beta - x^\beta \leq \hat{x}^\beta - \frac{x}{\hat{x}} \hat{x}^\beta = \hat{x}^{\beta-1} (\hat{x} - x).$$

Plug into (22):

$$\begin{aligned} \hat{x} - x_{t+1} &= (\hat{x} - x_t) - \frac{\hat{x}^\beta - x_t^\beta}{1 + \hat{x}^\beta} \\ &\geq (\hat{x} - x_t) - \frac{\hat{x}^{\beta-1} (\hat{x} - x_t)}{1 + \hat{x}^\beta} \\ &= (\hat{x} - x_t) \left(1 - \frac{\hat{x}^{\beta-1}}{1 + \hat{x}^\beta}\right) \\ &= (\hat{x} - x_t) \frac{1 + \hat{x}^\beta - \hat{x}^{\beta-1}}{1 + \hat{x}^\beta} \end{aligned}$$

Hence, if $1 + \hat{x}^\beta - \hat{x}^{\beta-1} > 0$, then whenever $x_t < \hat{x}$ we have $\hat{x} - x_{t+1} > 0$, i.e. the sequence stays below $\hat{x}$ forever. When $x_t < \hat{x}$, from (21) then we have $x_{t+1} > x_t$. Thus, the sequence is strictly increasing and bounded above by $\hat{x}$. By monotone convergence theorem, it converges to some limit $\leq \hat{x}$. Since $x_t \mapsto r_A^t(x_t)$ is strictly increasing, the only possible limit is $\hat{x}$ because $\hat{x}$ is the unique fixed point of (20) in $[0, \infty)$, so

$$x_t \uparrow \hat{x} \quad \text{and} \quad x_t < \hat{x} \quad \forall t,$$

i.e., no finite $\hat{t}$ with $x_{\hat{t}} \geq \hat{x}$ exists and $\lim_{T \rightarrow \infty} x_t = \hat{x}$, i.e., $\lim_{T \rightarrow \infty} r_A^T = \delta$. Here, the condition $1 + \hat{x}^\beta - \hat{x}^{\beta-1} > 0$ can be explicitly written as:

$$\left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} + \left(\frac{\alpha}{r_R - \delta} - 1\right)^{(1-\beta)/\beta} > 1.$$

In contrast, if $1 + \hat{x}^\beta - \hat{x}^{\beta-1} \leq 0$, then already at $t = 0$ we have $x_1 = \frac{\hat{x}^\beta}{1 + \hat{x}^\beta} \geq \hat{x}$, so crossing happens immediately. Note that $\hat{x}^{\beta-1} \geq 1 + \hat{x}^\beta$ is exactly the condition $x_1 \geq \hat{x}$. From (20) we know that $x_t = x_1 = \frac{\hat{x}^\beta}{1 + \hat{x}^\beta} \geq \hat{x}$ for all $t \in \{2, 3, \dots, T\}$. As a result,

$$r_A^t = r_R - \frac{\alpha}{1 + x_t^\beta} = r_R - \frac{\alpha}{1 + \left(1 - \frac{r_R - \delta}{\alpha}\right)^\beta}, \quad \forall t \in \{1, 2, \dots, T\}.$$

Moreover, the condition $1 + \hat{x}^\beta - \hat{x}^{\beta-1} \leq 0$ can be explicitly written as:

$$\left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} + \left(\frac{\alpha}{r_R - \delta} - 1\right)^{(1-\beta)/\beta} \leq 1.$$

C.3.8 Proof of Proposition B.12. Based on Lemma B.9, the fastest AI learning rate leads to $p_H^t = 1$ for all $t \in \{0, 1, \dots, T\}$. As a result, we have

$$\tilde{g}(x_t, p_t) = \begin{cases} 1 + \left[\frac{\delta - r_H}{r_R - r_A^t} \right]_+, & \text{if } r_A^t \leq r_H, \\ \left[\frac{\delta - r_H}{r_R - r_A^t} \right]_+, & \text{if } r_A^t > r_H. \end{cases}$$

Thus, the state transition (12) can be lower bounded by

$$x_{t+1} = x_t + \tilde{g}(x_t, p_t) \geq x_t + \max\left\{\frac{\delta - r_H}{r_R - r_A^t}, 0\right\} = x_t + \frac{\delta - r_H}{\alpha}(1 + x_t^\beta) \geq x_t + \frac{\delta - r_H}{\alpha}.$$

From $x_0 = 0$, by induction we have $x_t \geq \frac{\delta - r_H}{\alpha}t$. When $r_A^t = \delta$, from (11) we have $x_t = \left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta}$.

Let $\hat{x} \triangleq \left(\frac{\alpha}{r_R - \delta} - 1\right)^{1/\beta} > 0$.

Since $\delta > r_H$, for any finite target $\hat{x}$, choosing $t \geq \lceil \frac{\hat{x}\alpha}{\delta - r_H} \rceil$ guarantees $x_t \geq \hat{x}$ and therefore $r_A^t \geq \delta$. Thus, there always exists a finite stage $\hat{t} = \lceil \frac{\hat{x}\alpha}{\delta - r_H} \rceil < \infty$ such that $r_A^{\hat{t}} \geq \delta$. Moreover, it is not hard to see that $\lim_{T \rightarrow \infty} x_T = \infty$ and, consequently, $\lim_{T \rightarrow \infty} r_A^T = r_R$.

D Experiment Configuration and Additional Results

Core (Single-stage) Model. The model parameters of the core (single-stage) model (1) and (2) are listed as follows: $r_H = 0.80$, $r_R = 1.00$, $r_A = 0.40$, $c_H = 1.00$, $c_R = 0.60$; $\delta \in (0, 1.00)$, $b \in (0, 1.20]$. Fig. 3, Fig. 4, and Fig. 9 are generated using these parameters.

Multi-stage Model. The model parameters of the multi-stage model (15) are summarized as follows: $T = 300$, $\alpha = 0.60$, $\beta = 0.20$, $r_H = 0.80$, $r_R = 1.00$, $c_H = 1.00$, $c_R = 0.60$; $\delta \in (0, 1.00)$, $b \in (0, 1.20]$. Fig. 5, Fig. 6, Fig. 9, Fig. 10, and Fig. 11 are derived under this parameter setting.

Here we introduce the numerical settings to implement the Bellman recursion (16): (i) State discretization: We discretize the state x using a uniform grid on $[0, T]$. The value function $V_t(x)$ is represented at each grid point. In the experiments, we used 801 grid points in total for $T = 300$.

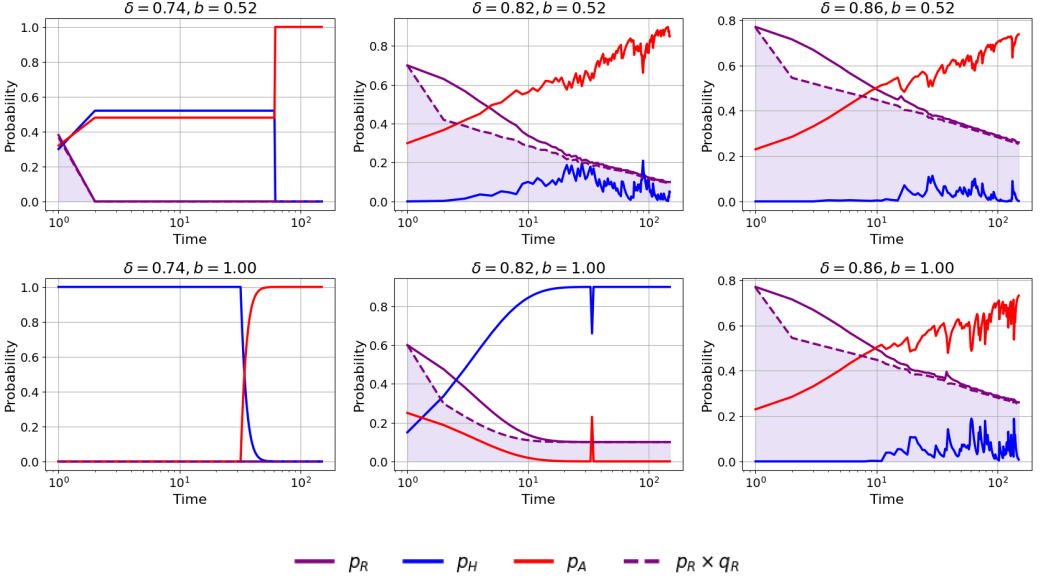


Fig. 10. Temporal view for equilibrium policy in the multi-stage model.

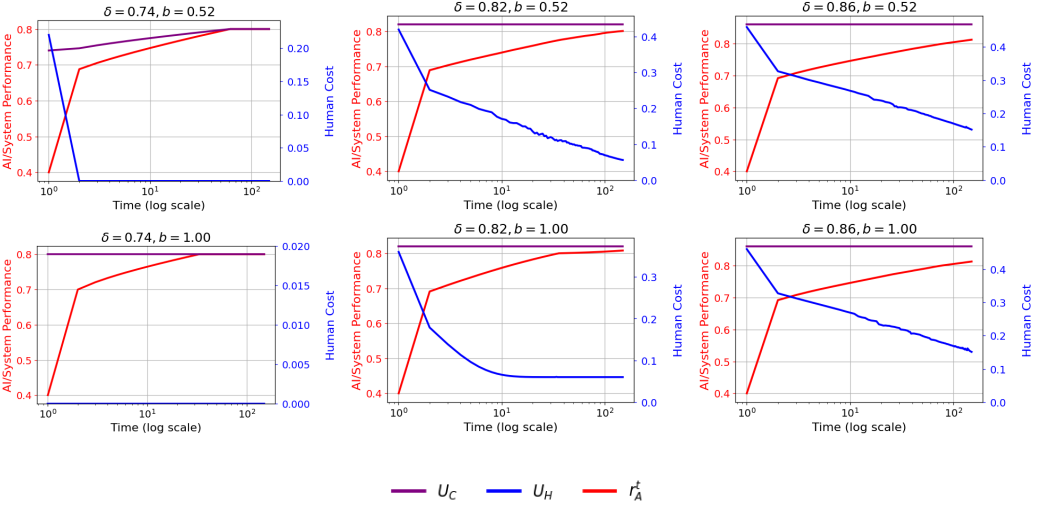


Fig. 11. Temporal view for performance and cost in the multi-stage model.

(ii) Value function approximation: For off-grid state x' , the continuation value is approximated by linear interpolation using the nearest on-grid states x_k and x_{k+1} :

$$V_{t+1}(x') \approx (1 - w) V_{t+1}(x_k) + w V_{t+1}(x_{k+1}),$$

where

$$w = \frac{x' - x_k}{x_{k+1} - x_k}, \quad x_k \leq x' \leq x_{k+1}.$$

- (iii) Action discretization: Similar to the state discretization, we discretize the continuous action space (p_H^t, p_R^t, p_A^t) . In particular, we enumerate a mesh of (p_H, p_R) pairs on a step size (default 0.01), and set $p_A = 1 - p_H - p_R$. For computational tractability, this approximation simplifies the procedure of solving a nonlinear program at each iteration to grid search over the feasible region.
- (iv) Numerical tolerance: The tolerance of numerical error is set to be 10^{-12} throughout.

Additional Results. Fig. 10 and Fig. 11 present the evolution of equilibrium policy and performance/cost, respectively, over time in the multi-stage model. From Fig. 10 we can observe that: (i) If we compare the subplots vertically, for fixed performance requirement δ , as the human budget b increases, the probability to use human-only decision making p_H also increases; (ii) Comparing the subplots horizontally, for fixed human budget b , the probability to use assisted decision making p_R and human-AI collaboration $p_R \times q_R$ increase as the performance requirement δ increases.

Similarly, we can see from Fig. 11 that: (i) Comparing the subplots vertically, for fixed performance requirement δ , if the human budget b increases, the coordinator's expected reward U_C increases while the human cost U_H decreases; (ii) If we compare the subplots horizontally, for fixed human budget b , as the performance requirement δ increases, both the coordinator's expected reward U_C and the human cost U_H increase.